QuantiTrack

Version 1.0

2026

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System Requirements

QuantiTrack has been optimized to run in MATLAB R2023a, and has also been tested with R2024a, R2024b, and R2025a. Some functions may not work in early versions. The following MATLAB toolboxes are required:

- Image Processing Toolbox
- Optimization Toolbox
- Statistics Toolbox

The following MATLAB toolboxes are optional:

- Parallel Computing Toolbox

Installation

Local MATLAB

1 - Unpack the zipped folder “QuantiTrack” into your matlab files folder (example (C:\Users\username\Documents\MATLAB\))

2 - Open Matlab

3 – Change the current directory to the QuantiTrack folder

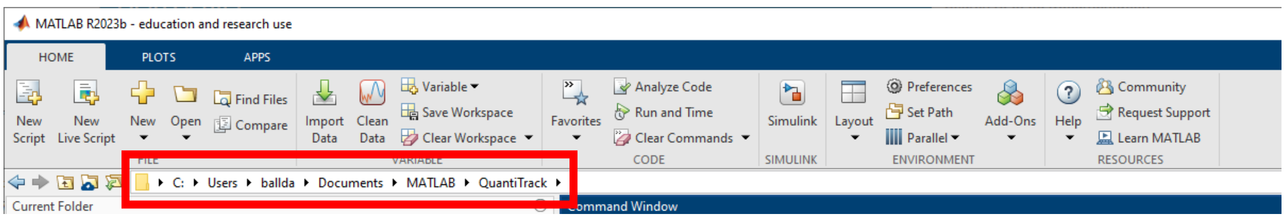

Figure 1: Change MATLAB working directory.

4 – Type **QuantiTrack_installer** at the MATLAB command prompt, and press enter. This will add the QuantiTrack folder and all sub-folders to the MATLAB search path. A progress bar will appear to let you know how much time is left. During installation, you will be prompted to enter a directory where all tracking results will be saved. It is recommended to create a new folder on a local drive and specify this as the Database location. This folder will subsequently be referred to as the “Track Database.”

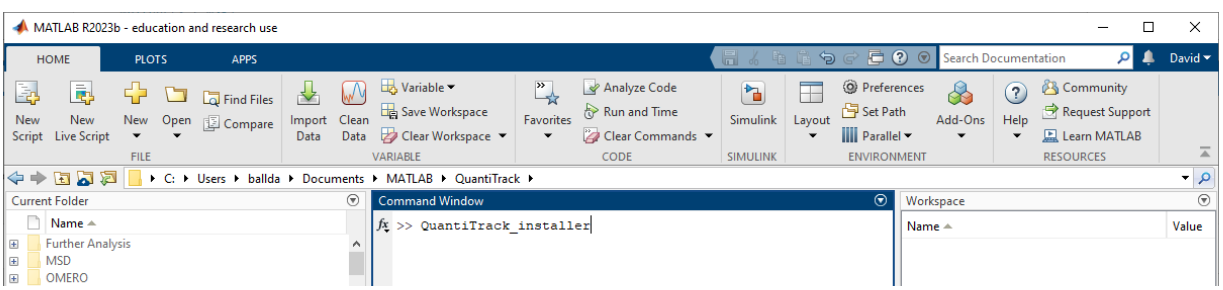

Figure 2: Run installer.

A local copy of default parameter values will also be created the first time QuantiTrack is run. This is to ensure that changed default values are not overwritten when updating the software.

MATLAB Online

QuantiTrack can be run entirely from a web browser through MATLAB Online (matlab.mathworks.com), with no local MATLAB installation required. This document walks a first-time user through loading the software, running the main tracking GUI on a test movie, and retrieving the results. The main analysis-through-tracking workflow is covered here; further analyses (SpotOn, pEMv2, Richardson-Lucy) run identically once a trackTable has been produced.

Prerequisites

- A MathWorks account (free at mathworks.com). Sign-in with a valid institutional license removes the Basic-tier session-time cap; without one, MATLAB Online is capped at 20 h/month.
- The following MathWorks toolboxes must be included in your MATLAB Online tier: Image Processing, Statistics and Machine Learning, Parallel Computing, Optimization, and Curve Fitting. Most campus licenses include all of these; the free Basic tier includes only the first two, so a free-tier user should upgrade to Basic+ or use a campus license before proceeding.
- A test movie (multi-page TIFF) uploaded to your MATLAB Drive. Any single-molecule fluorescence movie in .tif format works.

Steps

1. Sign in to MATLAB Online at https://matlab.mathworks.com with your MathWorks account. A new session launches the MATLAB Desktop in your browser — the layout matches that of the local desktop version (Command Window in the centre, Current Folder on the left, Workspace on the right).
2. Download the latest QuantiTrack release to your local machine. In the browser session's left-hand Current Folder pane, drag-and-drop the .zip onto MATLAB Drive to upload it. Right-click the uploaded file and choose “Extract”; MATLAB Drive unpacks it in place.
3. Alternatively, in the Command Window run:

```
!git clone https://github.com/dball07/QuantiTrack
```

which clones the repository directly into MATLAB Drive under ~/MATLAB Drive/QuantiTrack. Both methods produce the same layout.

4. Change into the QuantiTrack folder by double-clicking the QuantiTrack directory in the left pane and run the installer by typing the following in the Command Window:

```
QuantiTrack_installer
```

The installer adds every QuantiTrack subfolder to the MATLAB path with savepath, then prompts you to select a folder in which tracking output files (trackTables, per-movie .mat outputs, figures) will be saved. Choose or create a folder inside MATLAB Drive so the outputs persist between sessions.

5. Upload the test movie to MATLAB Drive: in the Current Folder pane, click the Upload button (or drag-and-drop the .tif file). Larger files can be uploaded via the MATLAB Drive Connector on a local machine, or streamed from cloud storage such as Google Drive or OneDrive via the corresponding MATLAB Drive integration.
6. Launch the main tracking GUI. In the Command Window:

```
QuantiTrack
```

The QuantiTrack App Designer window opens inside the browser. All interactive controls (file dialogs, threshold sliders, ROI drawing) behave as they do on a local desktop.

7. In the GUI, load your test movie, set the detection and linking parameters as described in the QuantiTrack User Guide, and run the tracking. Outputs (`trackTable_*.mat`, summary figures) are written to the folder you selected in step 4.
8. Retrieve the results. Right-click any output file in the Current Folder pane and choose “Download” to pull it to your local machine; whole folders can be downloaded as a .zip via the same menu. Results also remain in MATLAB Drive between sessions, so a later session can pick up where the previous one left off simply by re-launching QuantiTrack from the same folder.

Known limits

The Basic (free) MATLAB Online tier is capped at 20 hours of session time per calendar month and 5 GB of MATLAB Drive storage; a campus license removes the time cap and raises storage to 20 GB. Users processing many long movies should either sign in through a campus license or offload raw data to an external drive and stream in one movie at a time.

Parallel operations (parfor loops inside QuantiTrack) work on MATLAB Online but are limited to the number of workers granted by your tier: typically 4 on a campus license, versus the 8–16 workers available on a modern local workstation. Wall-time for parallel steps scales accordingly.

Tracking Workflow

Start QuantiTrack

1 – Type **QuantiTrack** at the MATLAB command prompt and press “Enter”.

Each time QuantiTrack is run, it will verify that the Track Database is available. If the specified path to the Track Database is invalid, you will be asked to specify the new path to it. If the Parallel Computing Toolbox is available, you will be asked if you want to use parallel processing. This choice can be changed in the menu “Tools → Use Parallel Computing”. The main graphical user interface (GUI) for the track analysis should open up (it should look like Figure 3).

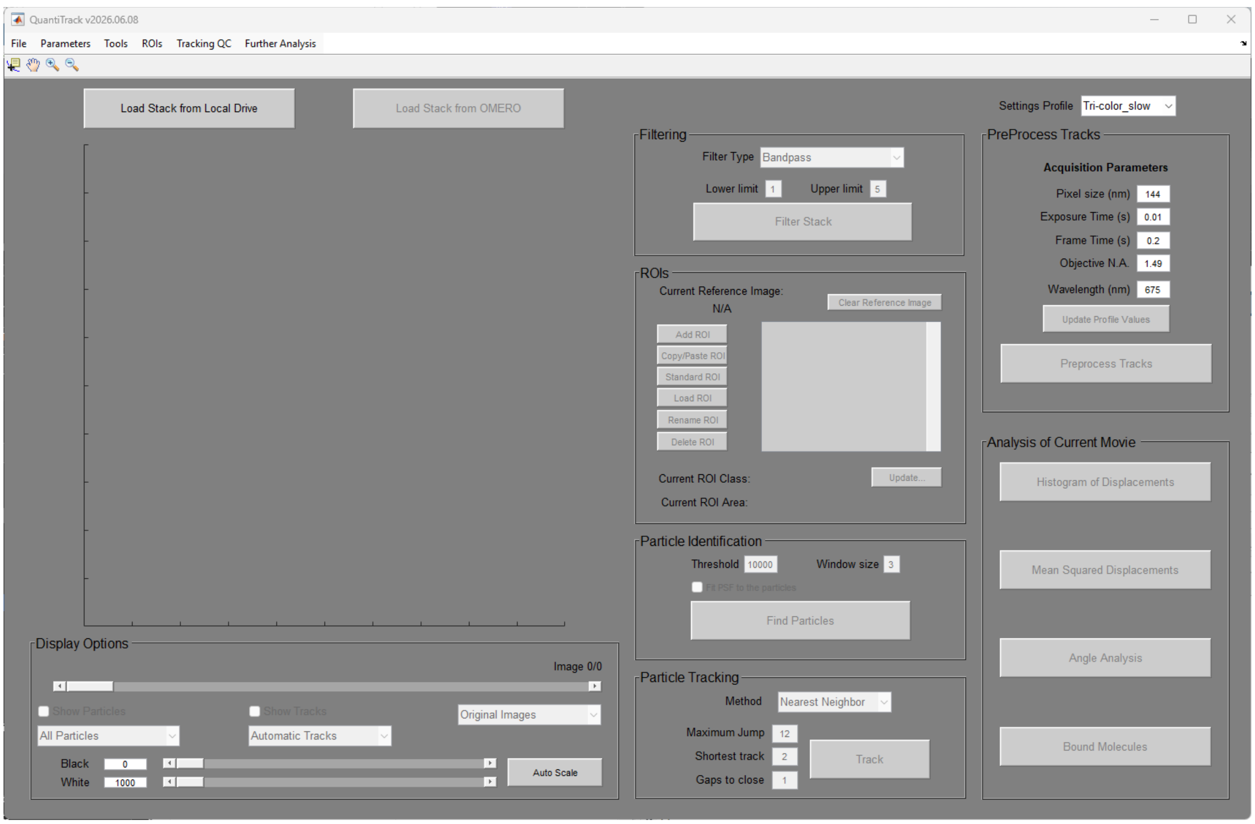

Figure 3: Main QuantiTrack Window

We can notice a series of buttons and some menus on top. The menus are used to perform some basic actions (load previously tracked data, save the tracked data, setting the parameters of the data acquisition and of the analysis) and some more sophisticated analysis (single molecule colocalization, and global analysis of multiple tracked movies). The main window allows us to perform the whole framework of the single molecule analysis on individual movies.

Load Movie

The first action to analyze a single molecule movie is to load the movie itself. You can do it by pressing the big button “Load Stack from Local Drive” at the top of the GUI. The movie needs to be in saved as a Tiff stack, i.e. one file containing all time-points, and it is preferable to have the files stored locally (not on some remote drive) for better performance of the routines. Once you select the movie you want to analyze, it should appear in the GUI (in the empty area on the left side of the GUI). You can scroll through the movie to assess its quality. If it looks like a bad movie (too many particles, blurred, lost focus etc.) proceed with the next movie by clicking again on the “Load Stack from Local Drive” button.

Set Parameters

Parameter settings are stored in Profiles, allowing multiple users and/or movies from different microscopes to be analyzed on the same computer. QuantiTrack opens with the last used Profile selected, and the current Profile used is shown in the dropdown menu in the top right of the Main window. Acquisition parameters for the current profile can be edited and saved directly below the dropdown. For control over more parameters, or to create a new profile, the Parameter Editor (Figure 4) can be used by clicking on the menu "Parameters → Edit Profile Settings" at the top of the window. With the parameter editor, one has control over all of the settings used by the specific tracking algorithms and down-stream analysis. The size of the pixel for our single-camera setup with the 150x objective the pixel size is 0.104 μm, while for the Tri-color scope with 100x objective, the pixel size is 0.144 μm.

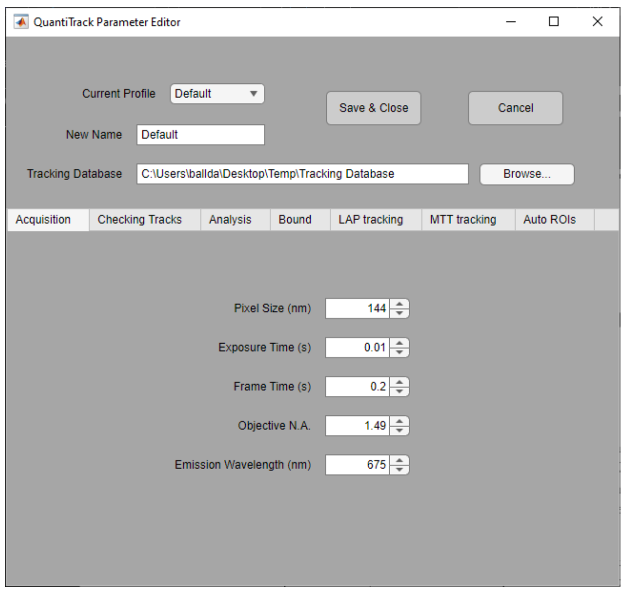

Figure 4: Parameter Editor

Filter Movie

The next step to detect the particles is to remove some of the noise by filtering the stack. We typically use a bandpass filter. The other option ('Local BG, no filtering (SLIMFast)') forces the Particle identification to be done by SLIMfast algorithm and will be discussed further below. In simple words the bandpass filter will smooth the image by "discarding" the features smaller than a certain threshold and larger than another threshold. You can change the thresholds by inputting these lower and higher limits in the corresponding boxes (Top of the center column of the GUI). For our system I find that a "Lower limit" equal to 1 and a "Upper limit" equal to 5 works well. You can then press the button "Filter Stack". As a result, you can switch between the original stack and the filtered stack by selecting the proper entry in the pop-up menu below the image window. NOTE: changing this entry will only affect what is visualized but NOT how the data analysis is performed. Tracking will always be done on the "Original Images".

Define Region(s) of Interest

Once we have processed the stack, we can select to analyze only the molecules found in a particular region, or regions, of interest (ROI) of the movie by pressing the button “Add ROI” in the center column. At this point, if desired, it is possible to open a reference image (for example a brightfield image of the same field of view) to identify the region of interest (for example a nucleus) more easily. Just click on the image to create vertices of a polygon surrounding the region of interest around the feature you are interested in, clicking on the first vertex or double-clicking closes the polygon and creates the ROI.

Identify Particles

The next step to track molecules is to identify the intensity peaks corresponding to the single molecules. To do so, we can set again two parameters. First, we can select a “Threshold” that allows discarding all those dim peaks that are probably due to noise. The value of this threshold depends on the brightness of your particles and on the amount of background you have. Select the threshold so that most of the peaks you think correspond to particles are detected, while keeping low the “false positive” particles. Next, we can select a “Window size” (in pixels). This basically reflects the minimum distance between two particles to be recognized as separated particles (I usually set it to the same value used for the higher limit of the bandpass filter). After setting these parameters you can hit the “Find Particles” button. The identified particles will be circled in red. Visually inspect the movie to check if you agree with the results of the particle identification procedure and change the threshold value accordingly (too many particles: increase the threshold - too few particles: decrease the threshold). Before proceeding with the next step, select the “Fit PSF to the particles” option and hit “Find Particles” again. This procedure is much slower but identifies the position of the particles much more precisely.

Track

Once you have found the particles, we need to link the different positions obtained at different frames as tracks. This is a very crucial step, as wrong tracking might result in completely nonsense results. QuantiTrack offers 3 different tracking algorithms: 1) Nearest Neighbor, which is a simple tracking algorithm that; 2) Linear Assignment Problem (LAP, uTrack); and 3) Multi-Target Tracking (MTT, SLIMfast). Nearest Neighbor tracking is the fastest of the three options, and works well for mostly stationary particles, but may fail for highly diffusive particles. It is recommended to test all 3 to determine which is best for your particular project.

After selecting the tracking method, the 3 key parameters should also be specified (For more control over LAP and MTT tracking their parameters can be further tuned using the Parameter Editor). First, you can set the maximum jump (in pixels) allowed between contiguous frames. The value of this parameter depends on the mobility of the studied molecule and the framerate of the movie acquisition. In case you're interested only in bound molecules, you can set the "maximum jump" relatively low (3 pixels), otherwise set it higher so that all the jumps performed by the particles are tracked. Keep in mind that if many particles are found, the routine might have trouble in connecting the particles correctly, especially if the "maximum jump" setting is too high. Therefore, if you want to track molecules moving long distances, acquire movies with a low number of particles (less expression, less fluorescent labeling ratio, etc.).

Then you can set the minimal length of the tracks, in order to discard tracks that are too short. I would keep this value low (1 or 2 frames). As we will discuss later there's another option to discard short tracks in subsequent analysis steps.

Finally, you can specify the "Gaps to close" option, which determines the maximum number of frames that a fluorophore can be dark and still be considered the same particle. This is useful when you have particles blinking, so that if "gaps" are found in the tracks they will be "filled".

After specifying all of these parameters, press the "Track" button. The algorithm might be challenged by two opposed situations (in both cases an error message will appear): Too many particles. You need to either increase the threshold to find the particles or set a lower "maximum jump" allowed. Too few particles, here the solution is to decrease the shortest track value.

Manually Check Tracks (optional)

Automatic algorithms for tracking aren't perfect and can produce artifacts. Therefore, it is advisable to manually inspect the tracked data one track at a time. This is a laborious process but can improve the fidelity of the tracking data. To make your checking less painful, you can discard very short tracks (they are not very informative anyway and they are more susceptible to errors). To discard short tracks, you can click on the menu "Parameters → Set checking parameters". I usually check only tracks that are three frames or longer, but it's up to you. This parameter can be saved in the Profile, so that you don't have to change it every time. To check the tracks click on the menu "Tracking QC → Manually Check Tracks" and an additional gui will open up. Here, you can check the tracks individually, edit the position of the particles, add or remove track points or delete individual tracks altogether. Once you're done checking all the tracks you can press the button "Done".

Preprocess Tracks

The next step is to pre-process the tracks, which converts position information from pixels to µm. Press the "Preprocess Tracks" to open up a new GUI (Figure 5)which will display in the top row: 1) all of the found tracks overlayed on the Sum-projection of the movie; 2) the distribution of the background around each detected particle (blue), and the intensity of each particle (red); 3) The distribution of jump distances within the tracks to verify the 'Max. Jump' parameter was adequate; bottom row: 4) the number of detected particles and tracked particles over time, which gives one an idea of the photobleaching present in the movie; 5) The distribution of the signal-to-noise ratio, with those values <1.5 highlighted in red (MAKE SURE THAT YOU ARE NOT ONLY TRACKING BACKGROUND, WHICH WOULD BE EVIDENT HERE WITH A LARGE PERCENTAGE <1.5); 6) Distance from each tracked particle to the 2nd closest particle in the subsequent frame, which gives an idea of over-crowding and therefore mis-tracking errors. There are several buttons on the right side of the GUI allowing you to perform operations (Make a movie of the tracks, copy, save figures, etc.). When finished, click "Done."

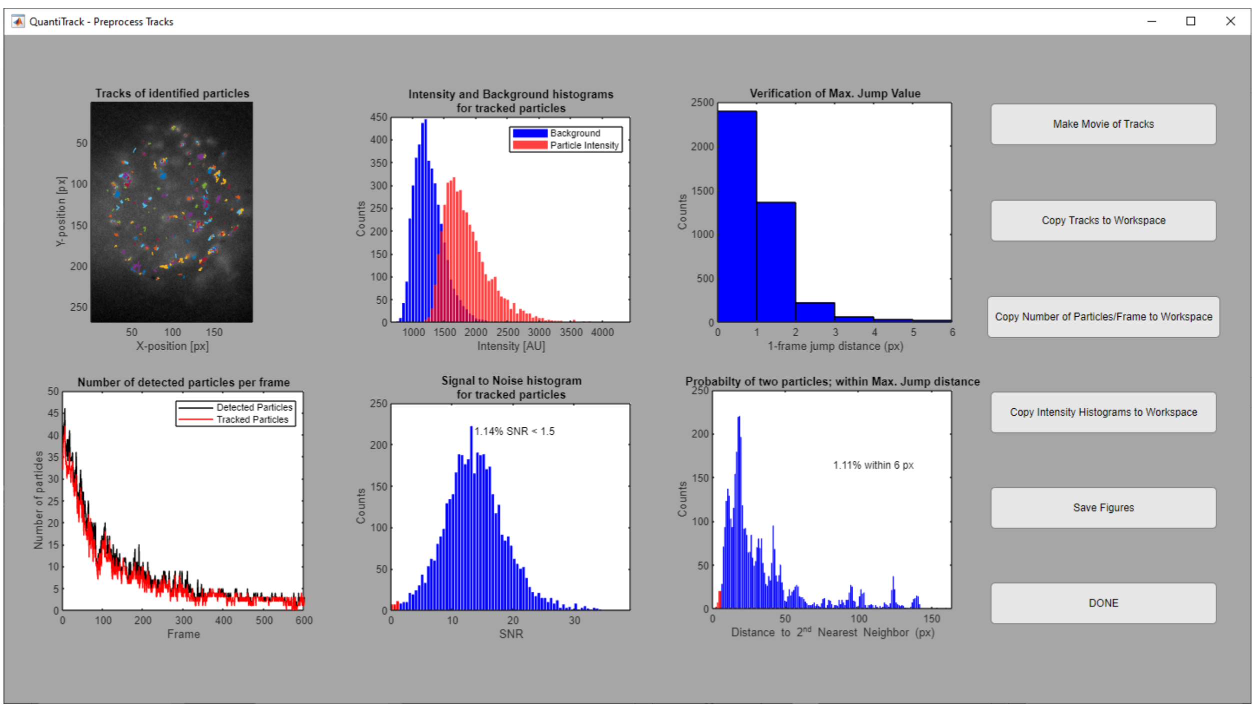

Figure 5: Preprocessing Window

Save Tracking File

At this stage it is worthwhile to save the work done. Select "File→Save to Track Database". The Track Database is organized into the following folder structure:

{Track Database} \ {Cell or Protein Name} \ {Condition} \ {Session}

For example, in Figure 6 the file will be saved in the directory C:\Users\ballda\Desktop\Temp\Track Database\H2B\10msExp_200msInt\2021-03-07. The first time saving a file to the database, you will be prompted to create a new folder for each of these.

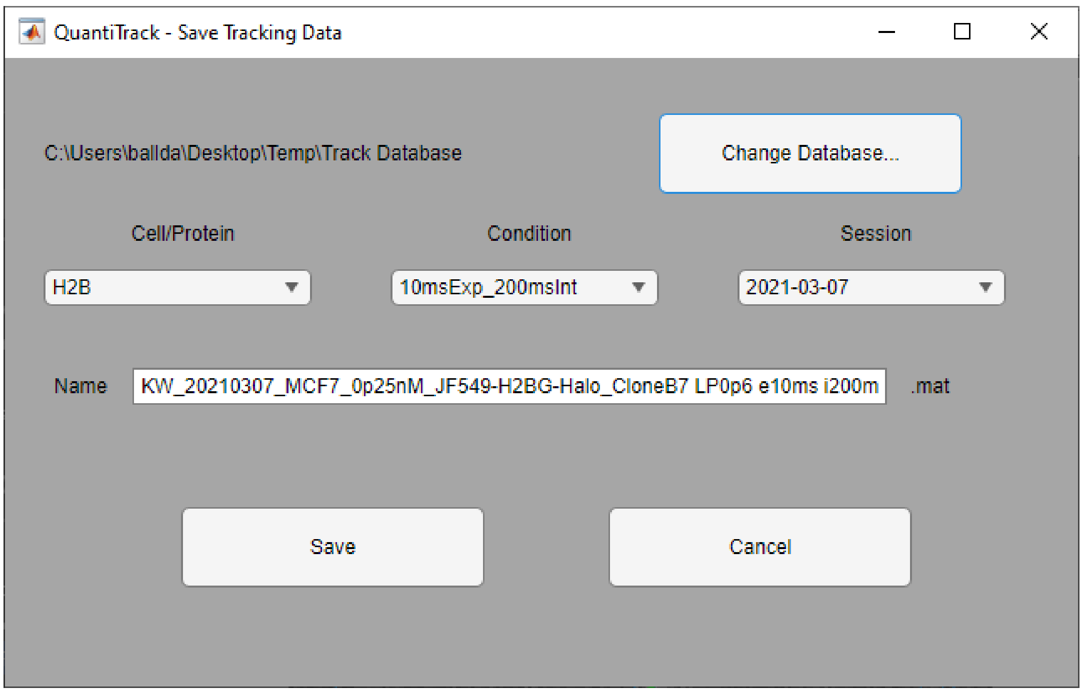

Figure 6: Save Tracking Data

When saving to the Track Database the tracking data is saved into a table format in addition to the .mat file that is saved. All data for a given Cell/Protein is stored in a single table, which simplifies and improves the speed when processing groups of files.

If you would also like to save the .mat file in a separate location, this can be done with the menu "File→Save a copy of the Track results".

Analysis of current movie

Once you processed the tracks you can start to analyze your data. You can either analyze individual movies or merge the tracks obtained for different movies and analyze the global behavior all-together. Here we discuss the analysis of the individual movies.

QuantiTrack offers four methods to analyze individual movies: analysis of the histogram of displacements; analysis of the mean squared displacements; analysis of angles (i.e. anisotropy); and analysis of bound molecules (i.e. residence times). These four tools are accessible from buttons in the main QuantiTrack window (Figure 7).

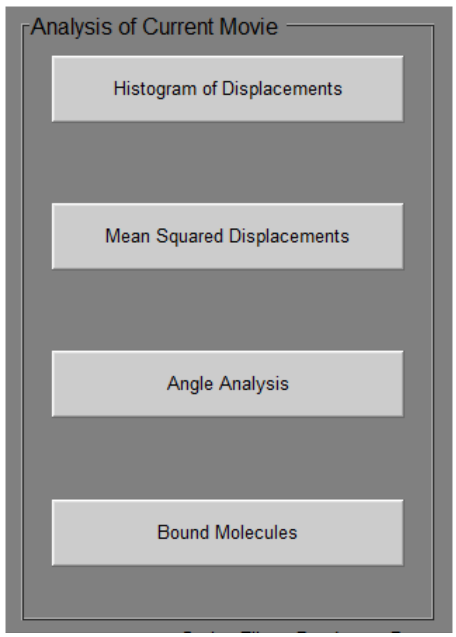

Figure 7: Analysis buttons in the QuantiTrack window

Histograms of displacements

This type of analysis quantifies the distances moved by individual molecules at different timescales, and fits it with models accounting for one, two or three diffusing components.

The first step in the analysis is to set some parameters. You can set these parameters in the Parameter Editor, or you can go directly to the analysis parameters by selecting "Set Analysis Parameters" from the "Parameters" menu in the QuantiTrack main window. There are two relevant parameters for the analysis of the histogram of displacements: The size of the bin of the histogram (I usually set it to be similar to the precision accuracy, ~ 0.03 um) and the number of time points that you want to measure (depends on the kinetics of your tracked molecules).

The first plot that shows when the processing is complete (Figure 8) is a "heat map" of the jumps ("time-dependent jump histogram"). The heat map represents how many molecules have jumped a certain distance (x-axis) in a certain time (y-axis). "Red color" means higher counts (more particles jumping that distance in that time), "blue colors" are lower counts. This plot is very informative, as, for example can be used to identify whether bound molecules are present (they will appear as a red streak, at short jumps). The Bayesian Information Criterion (BIC) for the three models are displayed in the text box below the heat map. The model with the lowest BIC best describes the data.

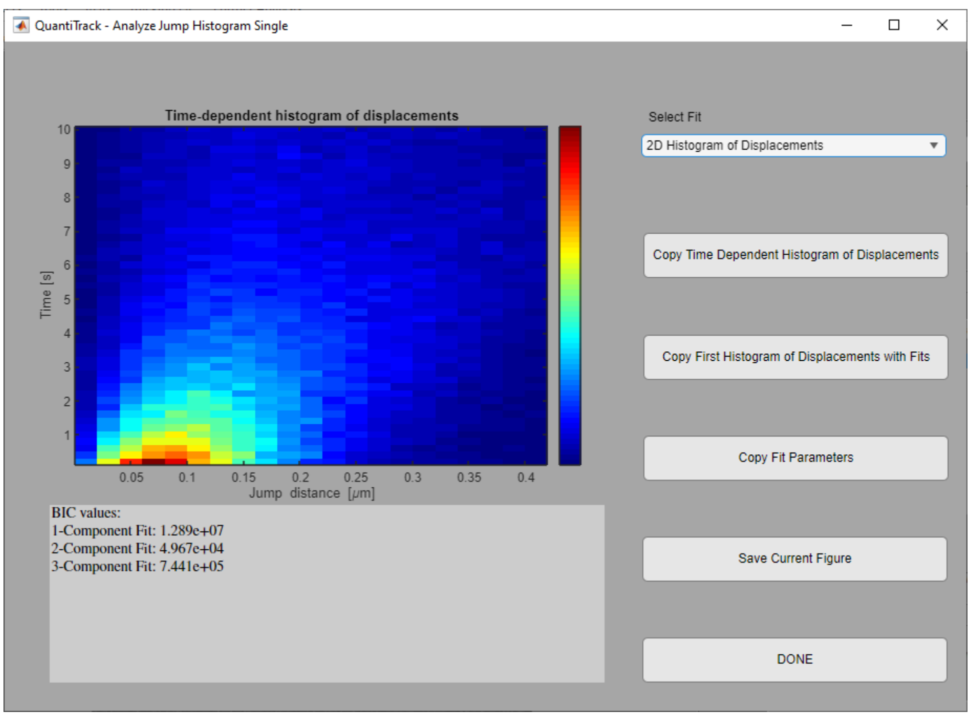

Figure 8: Analysis of Jumps

From the Jump Analysis window, you can inspect the comparison of the models' fit to the data with the dropdown menu. NOTE THAT THE FITS ARE BASED ON THE SINGLE-FRAME

JUMPS, I.E. THE FIRST ROW IN THE HEAT MAP. The data in Figure 8, for example, is best-fit with a 2-component diffusion model (Figure 9). You can copy the histograms, the fits and the fit results to the MATLAB workspace in table format, which can then be copied and pasted into Excel.

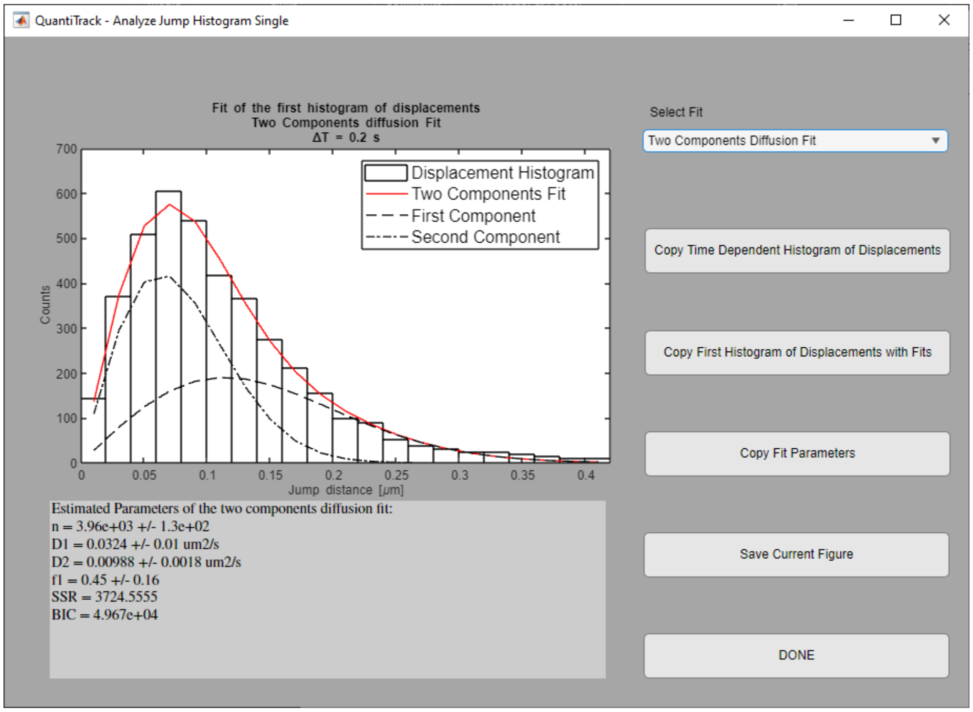

Figure 9: Fit of Single-frame jumps to a Two-component Diffusion Model

Mean squared displacements

Plotting the mean squared displacement (MSD) over time is another useful way to analyze particle motion. The MSD curve can be fit to determine whether the molecules are undergoing brownian, directed, or confined motion. Figure 10 shows the MSD analysis window. The MSD along with errorbars are shown in blue.

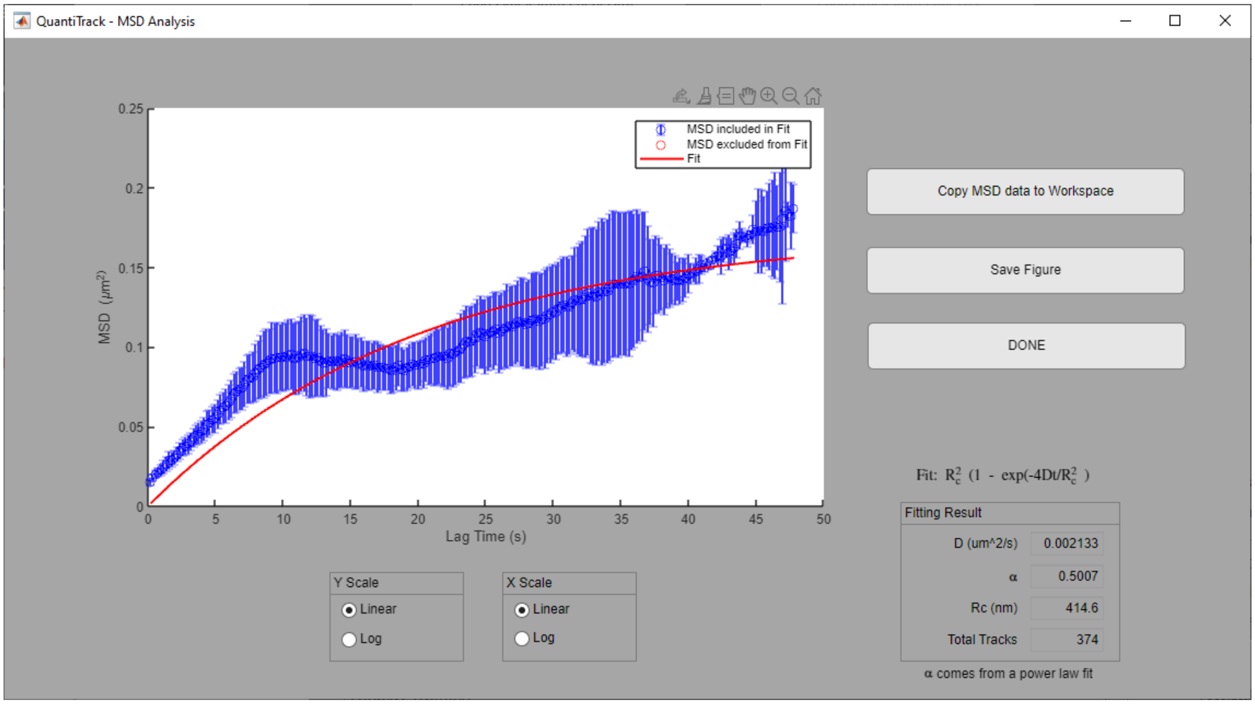

Figure 10: MSD Analysis, Single movie

The MSD analysis included in QuantiTrack fits the data in 2 separate ways. First, the entire curve is fit to a power law growth, with the resulting exponent, α, being the slope in log-log space. A slope of one would indicate brownian diffusion, while greater that one would show super-diffusion, and less than one would indicate sub-diffusive motion. The second fit is to a model of brownian motion confined to a region with a radius of R_c,

$$MSD(t) = R_c^2\left(1 - \exp(\frac{-4Dt}{R_c^2}\right)$$

For this fit, only points in the curve that have a standard deviation less than half of the mean value (in Figure 10, all points fit this criterion). Due to the low sample numbers, this fit is usually unreliable for a single movie, and the fit of many movies together should only be used for any quantitative analysis.

Angle Analysis

This type of analysis can reveal directed motion in a range of time-, and length-scales, and is based on the method described in Izeddin, et al.[1] It is run by clicking on the "Angle Analysis" button in the main QuantiTrack window. When the Angle analysis window opens, you will see 3 plots, the left of which is the distribution of angles formed by consecutive positions in all tracked molecules.

The middle, and right-most plots show the temporal and spatial anisotropy, which is defined as the base-two logarithm of the ratio of the number of angles in the forward direction ($-30° \leq \theta \leq 30°$) to the number of angles in the reverse direction ($150° \leq \theta \leq 210°$). To obtain the different timescales, the angles are calculated for integer multiples of a single time-step, i.e. the first value comes from consecutive time-points, the second value comes from every other time-point, etc. The spatial distribution is obtained by calculating the average displacement of the positions at the various timescales, which are then placed in bins of the specified width.

From the GUI, you can copy all of the angles and anisotropy data, copy the individual anisotropy curve data, and save the figure. You can also adjust the displayed plots by adjusting the minimum and maximum Time (or Displacement) and the bin widths. Once finished, click "DONE".

As with all of the current movie analyses, due to the small number of tracks used, the data shown here is quite noisy, and more reliable conclusions can be made by merging many movies and performing this analysis.

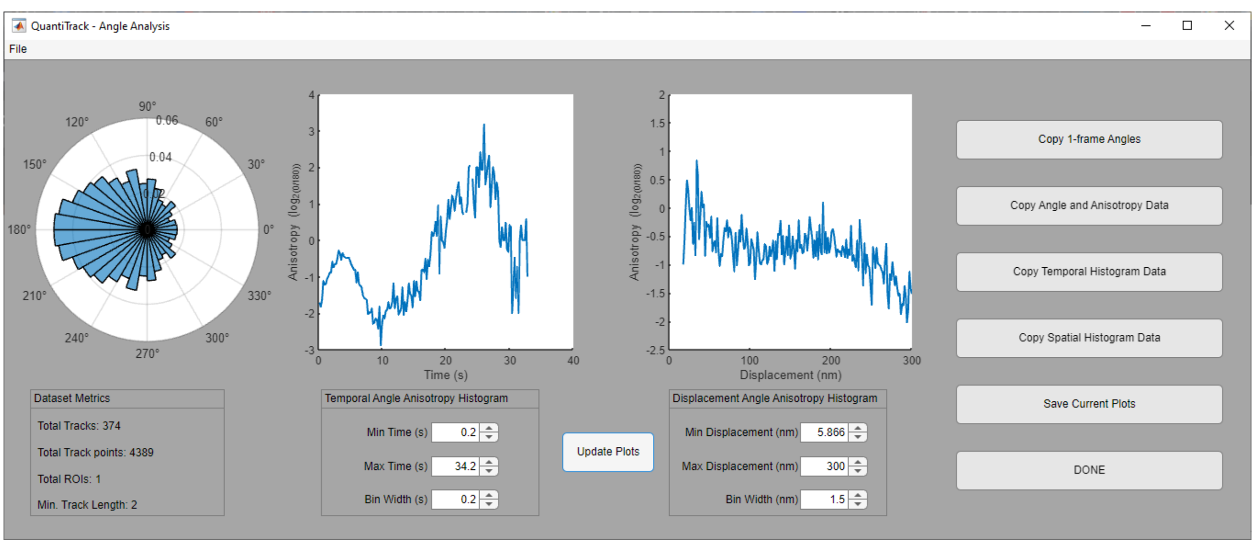

Figure 11:Angle Anisotropy analysis window.

Bound molecules

This kind of analysis is used to identify those track segments that reflect bound molecules and to display a histogram of the residence times (the time that each molecule spent bound).

Again, the first step to analyze the data is to set some parameters. You can set these parameters by selecting “Set Bound Parameters” in the “Parameters” menu, or opening the Parameter Editor and selecting the “Bound” tab. You can set three thresholds. The first one is maximum jump allowed from frame to frame for the bound molecules. The second a sort of “end to end” distance, the maximum distance you allow bound molecules to jump in a certain number of frames Nmin, specified below. The third parameter is Nmin itself, the minimum number of frames for which a molecule must display short displacements to be counted as bound.

After setting these parameters, you can run the analysis of the bound molecules by hitting on the button “Bound molecules”. Three plots will appear. The first is a plot of the tracks that have been identified as bound (green tracks) and the tracks that have been identified as free (black tracks). You can use this plot to double check that the “set bound” parameters have been properly selected. The second plot is the plot of the bound fraction over time, the bound fraction should appear rather constant, fluctuating around a mean value. The third plot is the histogram of the residence times. Again you can copy any of these variables on the clipboard.

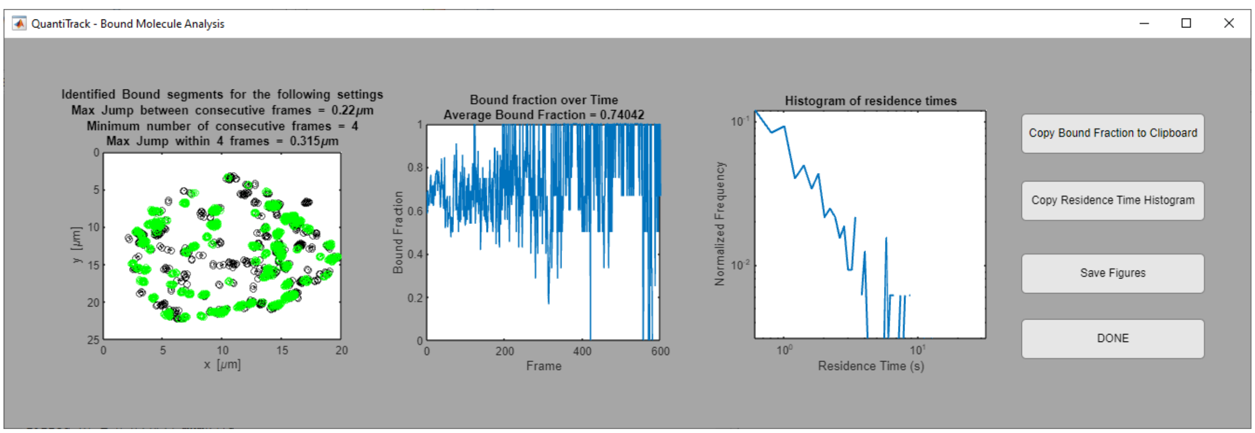

Figure 12: Analysis of bound molecules, current movie

Analysis of merged data

Typically, we analyze a group of tracked movies to obtain enough statistics to make conclusions and comparisons between different biological conditions. Datasets can be merged and analyzed in several ways. The methods currently available within the software include:

A) Jump Histogram Analysis

B) Residence time analysis with no photobleaching correction or correction based on

   a. Number of particles detected over time (Mazza, et al 2012)

   b. Tracking of a known stably bound molecule, such as histones (Garcia, et al 2021a)

C) Diffusion Coefficient analysis (Nguyen, et al 2021)

D) pEM analysis (Garcia, et al 2021b; Koo and Mochrie 2016)

E) Angle Analysis

F) Richardson-Lucy Analysis

Jump Histogram Analysis

Description

Kinetic modelling of the time-dependent distribution of displacements can reveal up to 2 mobility states within the tracking data. The analysis used here is the same as described by D. Mazza[2]. Spot-On is a similar type of analysis that has a web-based interface (https://spoton.berkeley.edu)[3].

Application of the Jump Histogram analysis to stationary molecules, such as histones, can also be used to find the appropriate parameters (r_{min}, N_{min}, and r_{max}) to specify bound molecules of more dynamic proteins.

Procedure

Perform the Tracking workflow on a number of movies (recommend at least 20 movies).

In the "Further Analysis" menu, select "Merge and analyze jump histograms

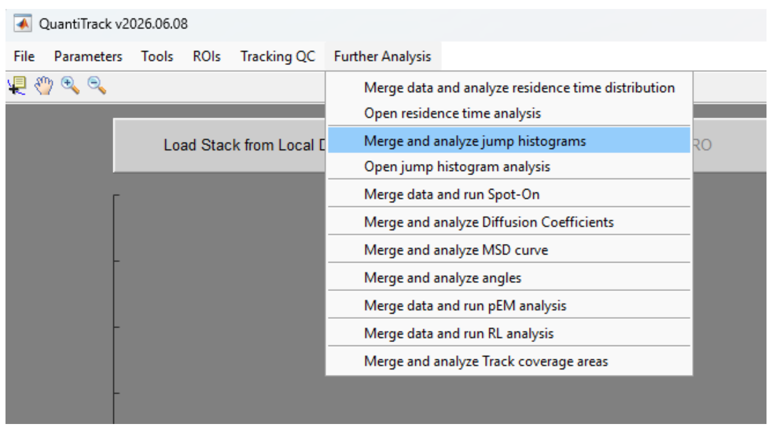

Figure 13: Further Analysis Menu showing "Merge and analyze jump histogram

Navigate to, and select, either a group of tracked .mat files, or a trackTable mat file.

Enter parameters:

Size of the histogram bin [um]: Bin size for the displacement histogram. This should be ~ half of the localization precision, so typically 0.01 µm is adequate.

Number of timepoints for the histogram: Number of frames over which to calculate the displacement histograms for fitting.

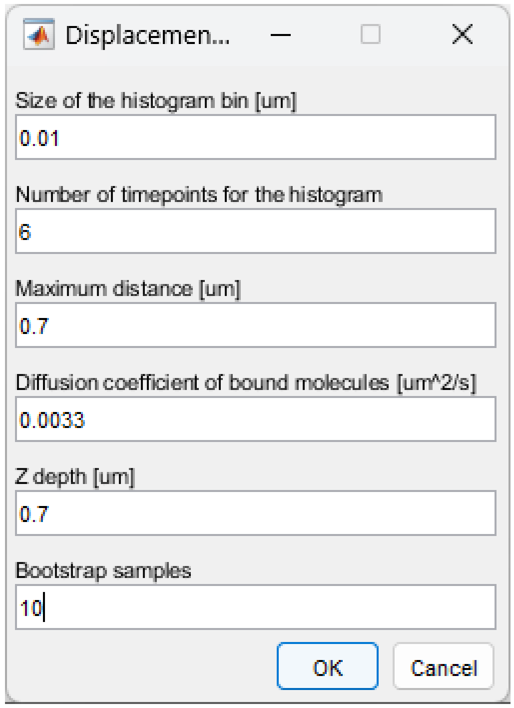

Figure 14: Parameters for the Jump Histogram analysis

You will then be asked what type of model to use for fitting, with the options of pure diffusion, Diffusion + binding, or 2-component diffusion + binding (Figure 15). You will be able to fit to the other models after the GUI opens.

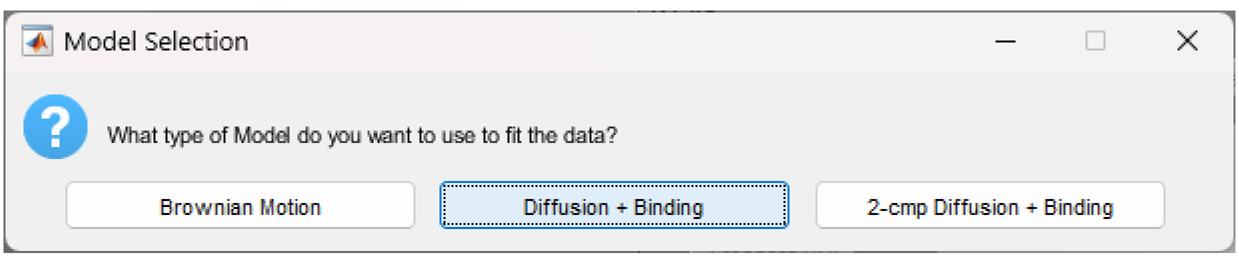

Figure 15: Model selection for Jump histogram fitting.

You will then be asked if you want to correct the histogram for photobleaching, which is only necessary if a large number of time-points (> 10) are used in the fit.

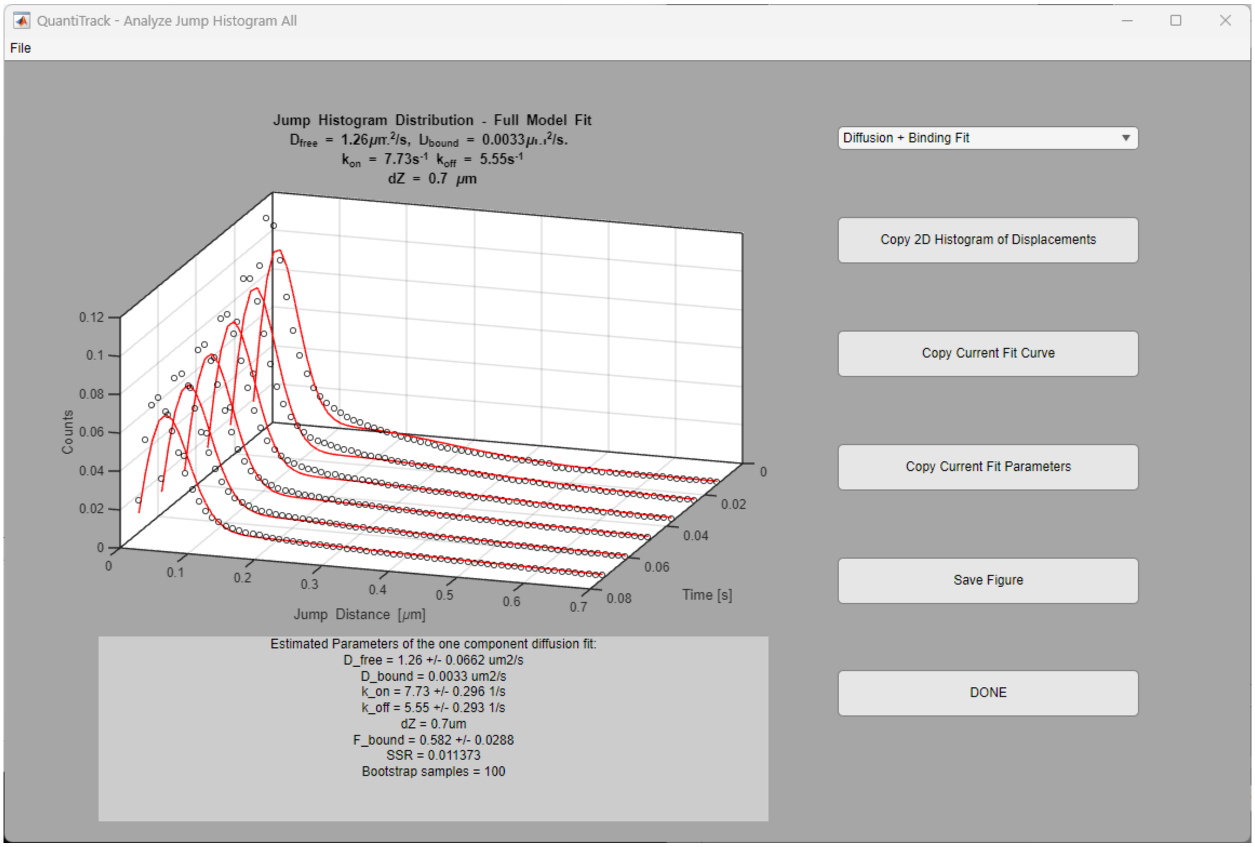

Figure 16: Jump Histogram Analysis GUI with heat map of jump distances

From the drop-down menu at the top, you can select to display the current fit or the heat map of the 2D histogram. Alternatively, you can choose to fit the data to one of the other models. After the fit is performed once, it will be displayed in the plot area as a 3D plot, and subsequent selections are displayed immediately.

Residence time analysis

Description

Analysis of the time that particles spend in immobile states can give insights into the binding kinetics of the molecule to its target. QuantiTrack generates the empirical cumulative distribution function (ECDF), or survival distribution, of the duration of immobile periods observed for a given molecule. The ECDF is then fit to several functions including 1) a single exponential decay 2) a bi-exponential decay 3) a power law decay, 4) a power law decay superimposed on a single exponential decay, and 5) an analytical equation.

While tracking molecules that are each tagged with a single fluorophore, disappearance of the particle during the movie from photobleaching can be confused for dissociation from its binding target. Ideally, the contribution to the survival distribution from photobleaching is removed before any curve-fitting is performed. QuantiTrack provides three options to correct the survival distribution for photobleaching: 1) the survival distribution of a stably bound protein, such as the histone H2B, or 2) by use of the number of detected particles over time, or 3) no photobleaching correction (Figure 17).

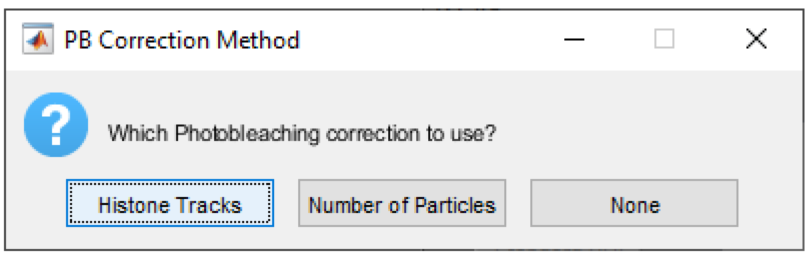

Figure 17: Methods for Photobleaching correction

By using the "Histone Tracks" method, the photobleaching correction comes from the survival curve obtained from tracking a stably bound molecule such as a histone. The method fits the histone's survival curve to a tri-exponential function. The slowest decay constant (k_{slow}) obtained from this fit is considered the photobleaching decay. The Survival curve from the protein of interest is then divided by a mono-exponential decay with the rate constant of k_{slow}.

Using the "Number of Particles" for photobleaching correction relies on counting the number of detected particles over time. The decay in the number of particles is used as a proxy for the photobleaching curve.

Preparation

Prior to analyzing the residence times for a protein of interest, it is important to determine how a truly bound molecule behaves, i.e. how far can it move in one frame (r_{min}), what is the minimum number of frames that it needs to move this distance (N_{min}), and what is the total distance moved after N_{min} frames (r_{max}). To accomplish this, a stably bound molecule can be tracked with the same conditions (laser power, exposure time, frame interval, etc.). We prefer to use histones for this purpose. When attempting to calculate bound parameters, the tracking should be performed such that we remove all diffusing molecules from the analysis. We therefore refer to this type of dataset as "Stable Histones."

In addition to the calculation of bound parameters, histones (or other stably bound molecules) can be used to estimate the photo-bleaching rate. For this type of photobleaching correction, it is also necessary to track histones with similar parameters to those used for the protein of interest. We subsequently will refer to this dataset as "All Histones."

Below we describe how to obtain these 2 datasets.

Stable Histones

Follow the instructions above for tracking a group of movies of labeled histones. For Particle Tracking, use the following settings to ensure only long, stable bound molecules are included:

Maximum Jump: 2

Shortest Track: 20

After pre-processing the tracks, save the tracking files in the Track Database under a condition called "Stable Histones".

Go to "Tools" and select "Calculate Bound Parameters from Stable Protein"

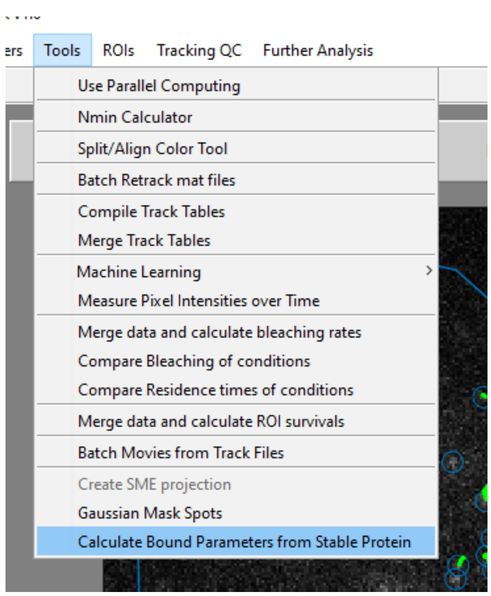

Figure 18: Calculation of Parameters to identify bound segments

Select all tracking files (or TrackTable file) that you would like to include in the analysis.

Enter parameters: The default parameters should be adequate for most applications, however, make sure the “Frame Time” parameter is correct. Probability Threshold is the chance that a diffusing molecule will be classified as bound.

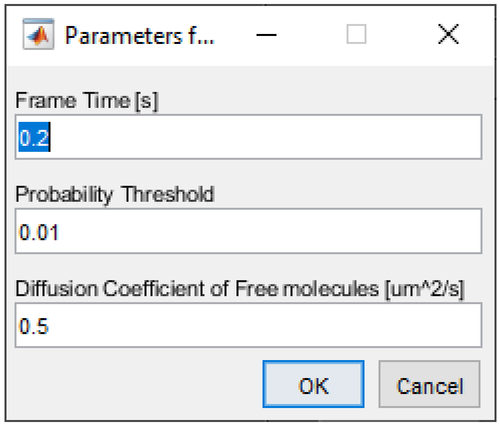

Figure 19: Variables used in the calculation of bound parameters

After the calculation is performed, a message box will appear stating that the Bound parameters have been updated. You can check the new parameters in the Profile Editor (Parameters-> Set Bound Thresholds).

All Histones

This is only necessary if using the Hitone tracks for photobleaching correction, if using the Number of particles, or no photobleaching correction then you are done with preparations

You could manually re-track the files you already tracked for the Stable Histones, but there is a better way:

In the "Tools" menu, click on "Batch Retrack mat files"

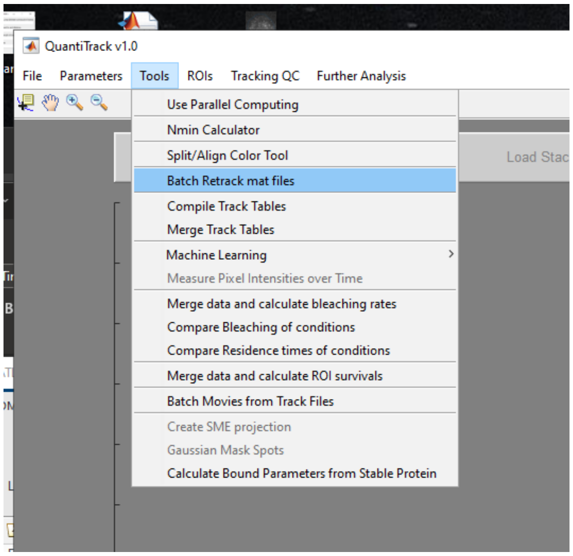

Figure 20: Location of Batch retracking option

Enter Tracking Parameters. It is best to set all of these to the desired value, such as shown below

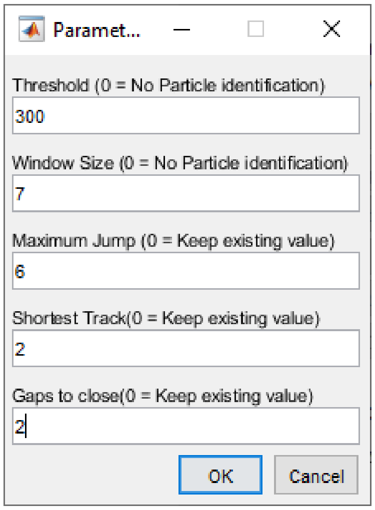

Figure 21: Parameters to use for retracking previously tracked files

Select the tracking files that you'd like to re-track (the files should all be in the same folder), or the trackTable mat file.

Select the save location within the tracking database. Tip: make this a new cell/protein folder so that the updated trackTable will not include both sets of histone tracking sets.

A progress bar should show up as its working. Once it's done, you are ready to proceed to the Residence Time analysis.

Procedure

After obtaining the parameters that specify bound segments and, if desired, tracking Histones for photobleaching correction, you are ready to analyze residence times.

In the "Further Analysis" menu, select "Merge data and analyze residence time distribution (Figure 22).

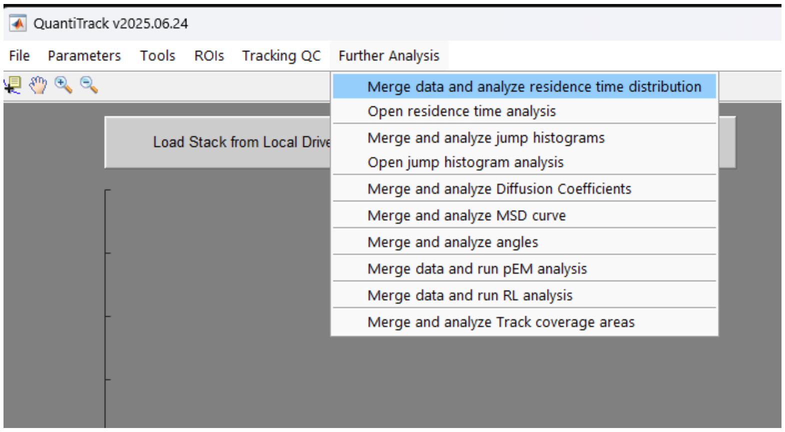

Figure 22: Residence Time Analysis location

Select the desired photobleaching correction method (Figure 17).

Choose the analysis parameters as shown in Figure 23. The Maximum jump between consecutive frames is r_{min}, Maximum end to end distance is r_{max}, and Minimum length of bound tracks is N_{min}. Also make sure that the Frame Time is correct. Because the survival probability at long lag times can be very noisy due to the low number of track points that it includes, there is also the option to only fit points in the survival that include a certain number of tracks.

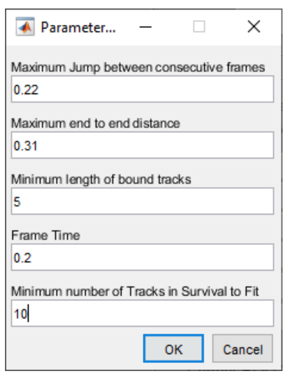

Figure 23: Parameters for Residence Time Analysis

After clicking "OK," the files will be read, and the residence times will be calculated, and the Analyze Residence Times GUI will appear. From this GUI, The residence time analysis can be saved from the "File" menu, or you can load in a previous analysis. In the "Tools" menu, you will find the "Generate Pie Chart", which creates a pie chart (for the best fit curve) showing the sub-populations that are diffusing, short bound (with mean residence time), and if applicable long bound (with mean residence time).

you can choose which fits are displayed (Figure 24, upper right), copy the data, analysis parameters, fit parameters, or save the figure.

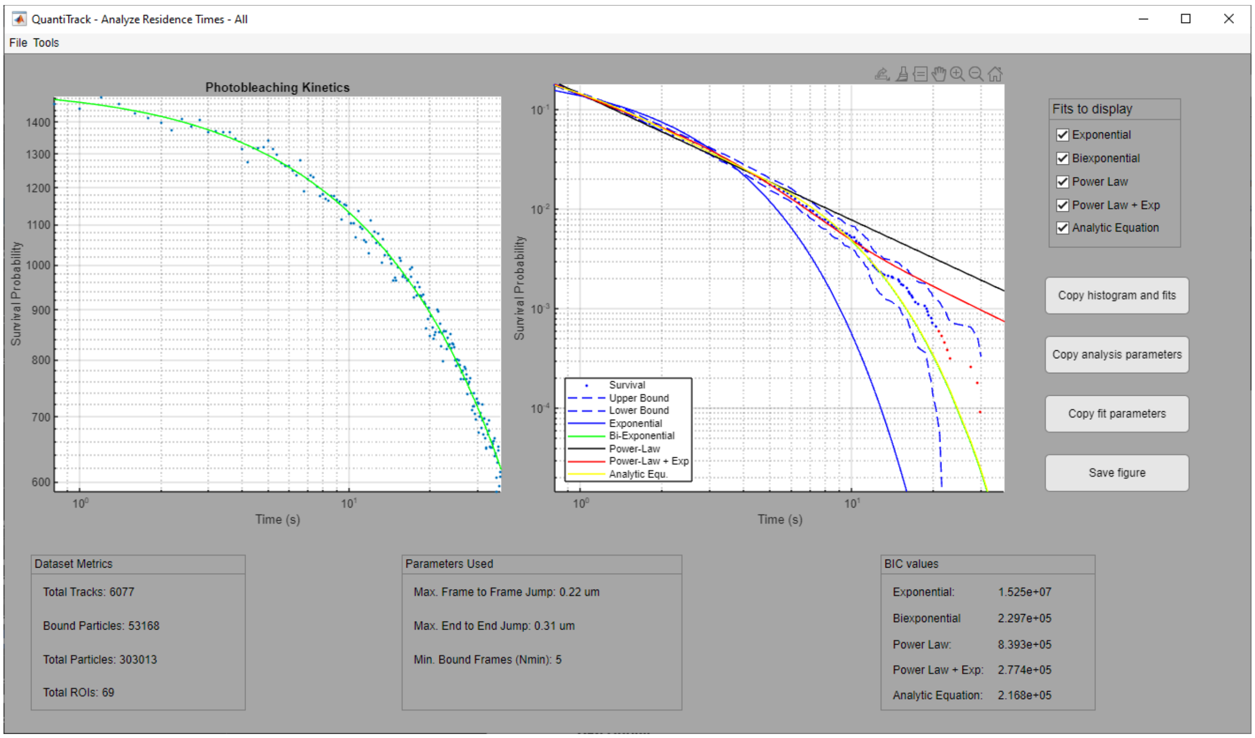

Figure 24: Analyze Residence Times Interface

At the bottom of the GUI, you can see certain metrics about the dataset (i.e. Number of tracks analyzed, number of bound particles detected, number of total particles, and the number of ROIs that were tracked), the parameters used for the analysis (r_{min}, r_{max}, N_{min}), and the Bayesian Information Criterion (BIC) values for the various fits. NOTE THAT THE LOWEST BIC INDICATES THE BEST FITTING MODEL.

When copying Histogram and fits, Analysis parameters, or Fit parameters using the buttons on the right side, the data is copied to the MATLAB workspace as a table structure.

Diffusion Coefficient Analysis

Description

This tool attempts to fit the MSD of individual tracks to obtain a diffusion coefficient (D_i), and produces a histogram of log(D_i). The histogram is fit to mixtures of normal distributions with up to 3 populations. The BIC and AIC of each fit is used to determine which is the best. This analysis is similar to the diffusion coefficient analysis described in Nguyen, et al 2021[4].

Procedure

In "Further Analysis", select "Merge data and analyze residence time distribution", then "Merge and analyze Diffusion Coefficients"

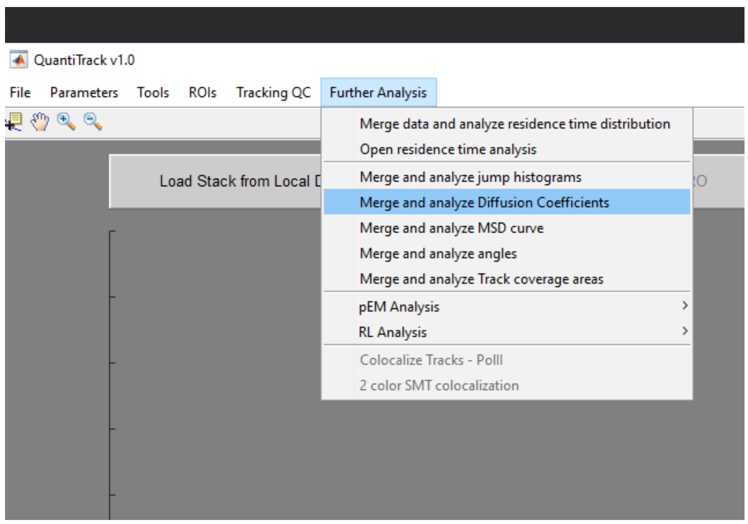

Figure 25: Diffusion Coefficient analysis

Set Parameters for construction of the histogram (i.e. bin limits (can be auto-scaled), number of bins), frame interval time, the maximum frame to include in the MSD calculation, and the minimum track length to include.

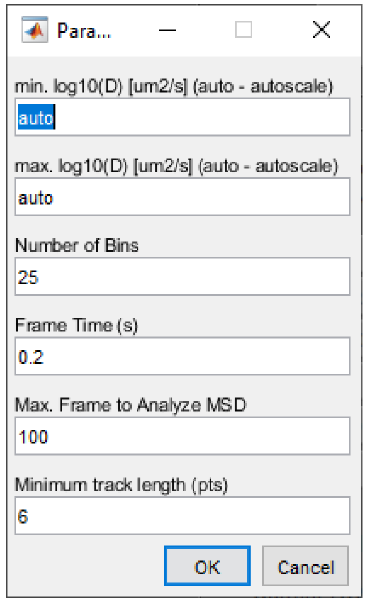

Figure 26: Parameters for Diffusion Coefficient analysis

Select the tacking file(s) that you would like to analyze.

A progress bar will give an indication of how long the processing will run as it reads in each Histone tracking file. If multiple ROIs were present in the tracking data, you will additional see a window popup asking to choose the ROI name to analyze, If "All" is chosen, then all tracks will be considered. If ROIs were divided into classes, it will first ask if you want to analyze based on ROI name or class.

The analysis GUI (Figure 27) will be displayed when the processing has completed. The graph shows the frequency of occurance of log(D_i) in blue bars, and the best fitting Gaussian mixture model in red with the individual populations in black. The statistics of the best fit are shown in the lower right of the GUI. The other fits can be inspected using the drop-down menu in the upper right. From here, you can save the figure, and/or the underlying data and fit parameters.

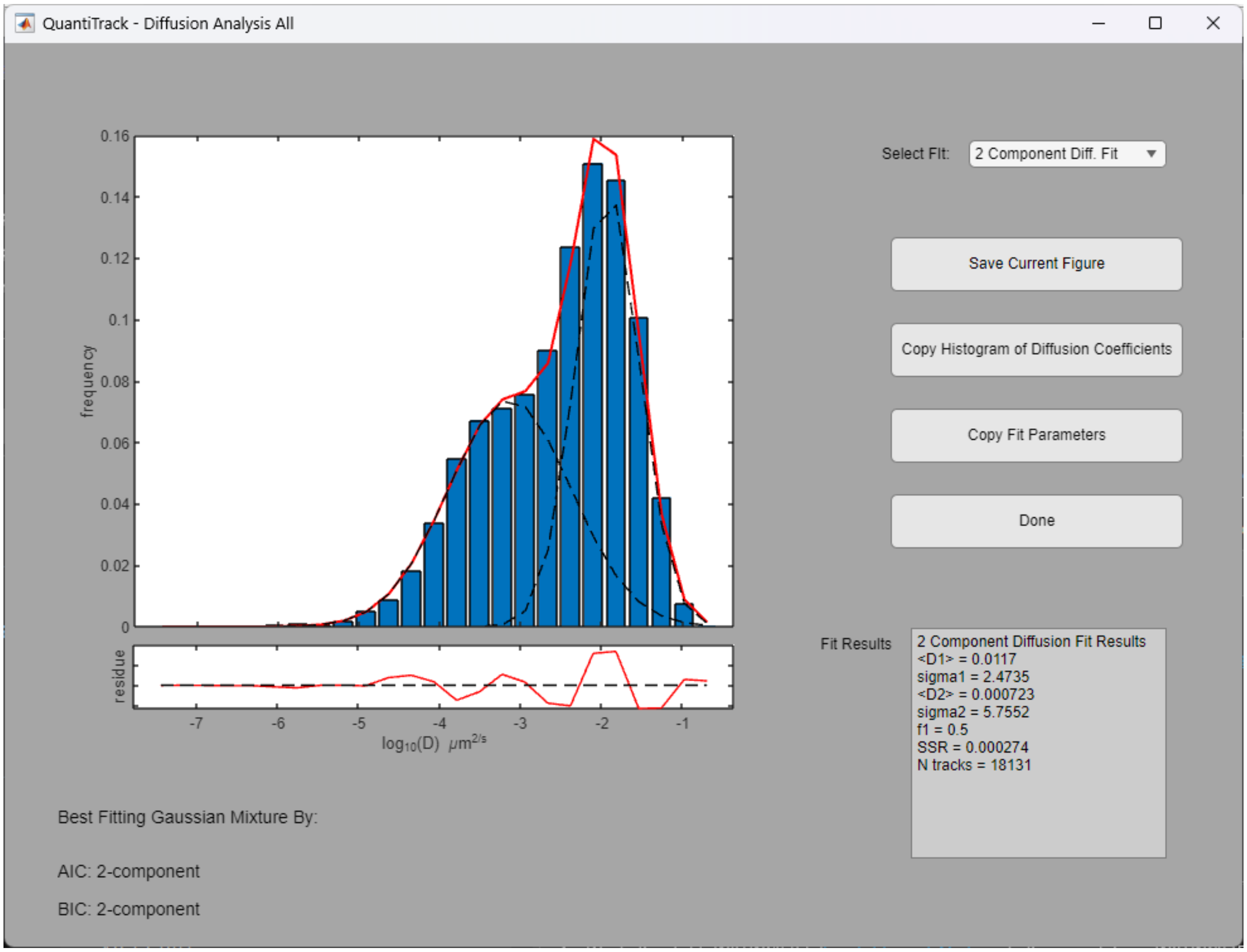

Figure 27:Distribution of Diffusion Coefficients window

pEM Analysis

Description

Perturbation Expectation Maximum (pEMv2) is a powerful tool that allows one to extract the various diffusion states that may be present in tracking data[5]. pEM analyzes track segments rather than entire tracks, as long tracks may include several diffusion states. Therefore, the length of splitting tracks is a critical parameter, and should be chosen in such a way that the probability of switching between states within a sub-track is minimized. In practice, we typically use a split length of 15 for fast imaging with time intervals ≤ 20 ms, and a split length of 7 for slower tracking with time intervals ~200 ms. When complete, the pEM analysis results in the creation of a number of folders and files which show the number of states, their proportions, and their diffusion characteristics (i.e. MSD curve, Diffusion coefficient, and Radius of confinement). A few steps are necessary to prepare the data prior to performing pEM analysis.

Preparation

The pEM analysis scripts require tracking data to be organized into trackTables that have specific column headings. If tracking data has been saved to the Track Database, then the appropriate trackTables should already exist. If working from tracking files saved in matTrack, the trackTables can be made by running a couple of functions within QuantiTrack. First, the files should be organized into the appropriate folder structure. See the section **Error! Reference source not found.** for details. Once in the proper folder structure, the trackTable can be created by following the instructions outlined in the section **Error! Reference source not found.**.

Procedure

Once you have a trackTable, pEM analysis can be started from the Further Analysis menu, by selecting "Merge data and run pEM Analysis (Figure 28).

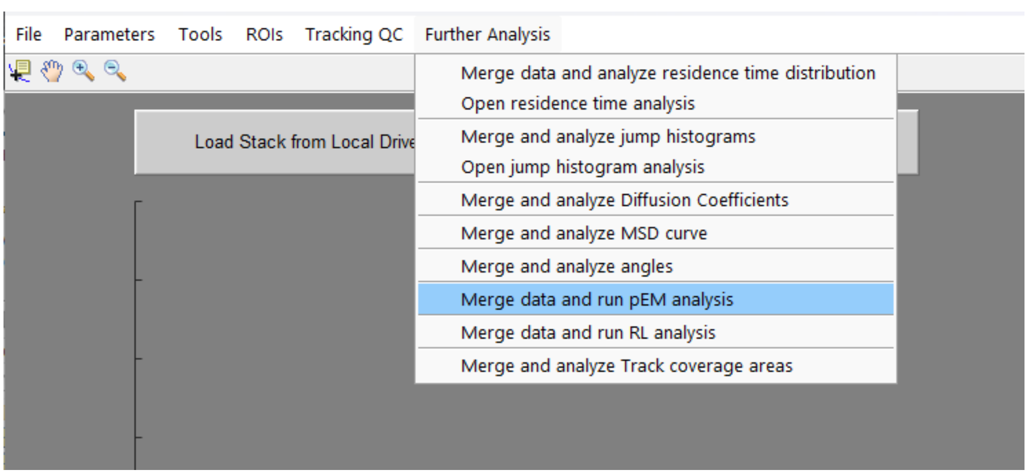

Figure 28: Menu for pEM Analysis

You will be asked to choose the trackTable to be analyzed, followed by some parameters (Figure 29). These parameters are:

1) **Split length (frames)**: length in frames of the sub-tracks, any track or sub-track less than the split length will be discarded. This should be chosen to minimize state transitions within a sub-track. As such this parameter will be highly dependent on the protein studied. We have had good results in a range of studies using a split length of 7 for movies collected with a time-lapse interval of 200 ms. For faster movies, time-lapse intervals ≤ 20 ms, we typically use 15 frames.
2) **Min number of states**: The minimum number of states that can be present in the data
3) **Max number of states**: The maximum number of states that can be present in the data.
4) **Features**: off-diagonal values from the covariance matrix to use for evaluating the sub-tracks. We typically use 3 features for short (7 frame) sub-tracks, and 5 features for longer (15 frame) sub-tracks.
5) **Reinitialization Trials**: number of times to re-try the expectation maximum. 20 should be sufficient for most experiments, but higher numbers may produce more consistent results across runs at the expense of computing time.
6) **Perturbation Trials**: number of perturbations to try. 200 should be sufficient for most experiments, but higher numbers may produce more consistent results across runs at the expense of computing time.
7) **Max iterations per trial**: Threshold after which if the model has not converged, the trial is aborted.
8) **Convergence criterion**: Difference in the model fit between the current iteration and the previous used to indicate convergence of the model.
9) **Min value of max posterior prob**: As pEM is a probabilistic method, each sub-track will have a probability to belong to each state. This parameter specifies the minimum probability that can be used to specify a properly classified sub-track. However, the next parameter has shown to be a better indicator, and this is usually set to 0.
10) **Min difference between posterior prob of best 2 states**: Minimum difference between the 2 highest probability states that would indicate that a sub-track is well classified. For example, with this parameter set to 0.2, if a dataset is found to contain 2 states, any sub-track must have a posterior probability for a state of ≥ 0.6, because this would mean the posterior probability of the other state is ≤ 0.4 and the difference between them would be at least 0.2. It is similar but more complicated for higher-order models.
11) **Points to fit MSD**: number of time-points used to fit the MSD curve to find the Diffusion coefficient. Because the motion is typically confined causing the MSD curve to plateau, only the first few, i.e. 3, points should be used for this calculation.

12) **Resamples used in Bootstrap**: If it is desired to obtain error bars on the state populations, this parameter can be set to a non-zero value.
13) **pEM runs to perform**: Due to its stochastic nature, several runs of pEM may produce slightly different results, and it is recommended to run it several times to verify the results. This parameter species the number of times to run pEM, producing a separate output folder for each run. If parallel processing is enabled, these runs will be run in parallel.

After verifying all of the parameters, click OK. The pEM analysis will begin, and a progress bar will appear. Depending on the computer used, the size of the dataset and the complexity of the model required to fit it, pEM may run for several hours before completing.

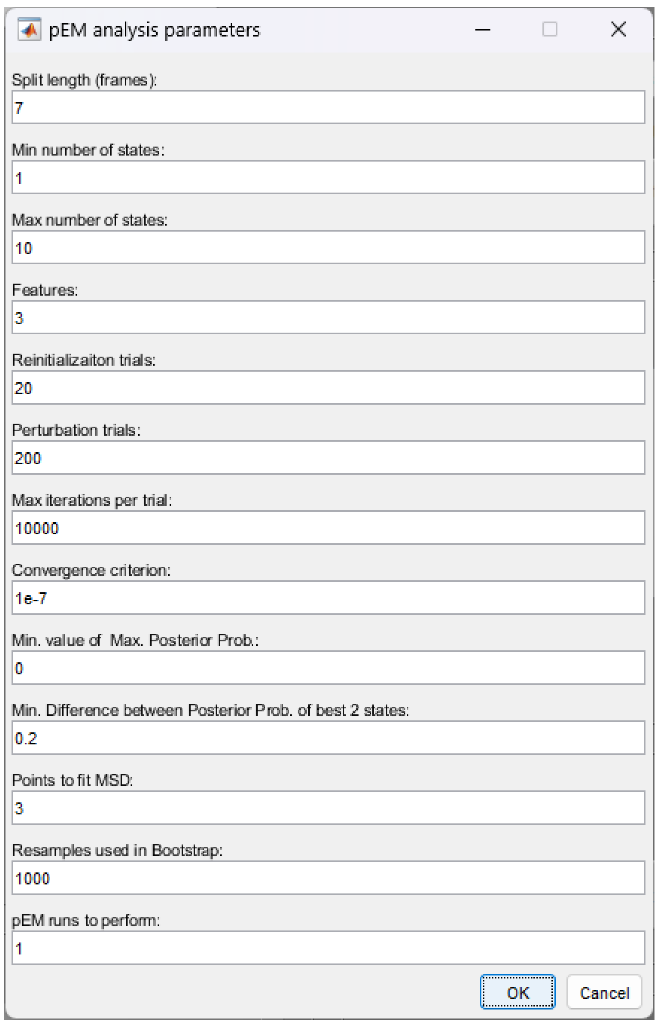

Figure 29: pEM parameter input

Output

Because of the time required to complete the pEM analysis, all of the outputs are automatically saved in the same folder as the 'trackTable' file was located in a folder(s) named 'pEM_results_DATESTRING'. If multiple runs were started at the same time, there will be an indication in the name.

Inside the pEM_results folder you will find .mat file that contains the original trackTable and the results of the pEM analysis in a table format. There is also an excel file called 'proportions_DATESTRING.xlsx' that contains the diffusion coefficients (OptimalD), Localization Error (OptimalS), and the proportion of the population in each state. There is also a folder called 'post_processing' that contains plots and tables for the analysis of all data in the trackTable as well as analysis of the data pooled by the conditions listed in the trackTable.

The condition folders contain plots (saved as matlab .fig files) of representative tracks from each state in the '*CONDITION_NAME*_tracks_*DATESTRING*.fig'. The file named '*CONDITION_NAME*_swarm_msd_cdf_popFrac_*DATESTRING*.fig' contains:

1) Swarm charts showing the distribution of the maximum posterior probability for each state.
2) Heat map showing the state of each subtrack within a track, which is helpful if looking for state transition probabilities.
3) Plots examining the length of tracks and how the state populations are distributed based on track length.
4) Cumulative distribution function for the Diffusion coefficients for each state.
5) plots of the MSD curves for the found states
6) Distribution of diffusion coefficients for the two least mobile states.
7) Stacked bar graph showing the state populations on a cell-by-cell basis

Angle Analysis

Description

This type of analysis can reveal directed motion in a range of time-, and length-scales, and is based on the method described in Izeddin, et al.[1] The analysis calculates the temporal and spatial anisotropies, which are defined as the base-two logarithm of the ratio of the number of angles in the forward direction (-30° ≤ θ ≤ 30°) to the number of angles in the reverse direction (150° ≤ θ ≤ 210°). To obtain the different timescales, the angles are calculated for integer multiples of a single time-step, i.e. the first value comes from consecutive time-points, the second value comes from every other time-point, etc. The spatial distribution is obtained by calculating the average displacement of the positions at the various timescales, which are then placed in bins of the specified width.

Procedure

From the "Further Analysis" menu, select "Merge and analyze angles" (Figure 30).

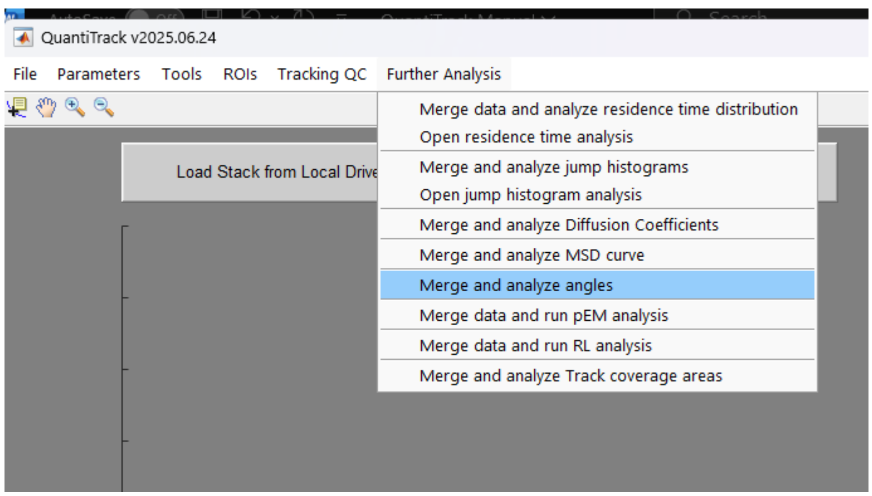

Figure 30: Running Angle Analysis

You will be asked to choose either a trackTable or a group of tracked files to be analyzed, followed by some parameters (Figure 31). These parameters are:

1) **Minimum track length (frames)**: Removes any tracks from the analysis that contain less than this number of time-points.
1) **Frame interval (s)**: Time between frames in the movie. This will be populated with the value used in the QuantiTrack main window.
2) **Bin size for time (s)**: Specifies width of bins when grouping angles based on time-lags of the jumps.

3) **Bin size for distance (nm)**: Specifies width of bins when grouping angles based on average distance of the jumps.
4) **Bootstrapping iterations**: number of random samples to take from the data with replacement to estimate the error in the measurements.
5) **Remove Angles with small steps**: Set to 1 to discard any angle that includes a displacement less than the localization precision of the movie. If set to 0, all displacements will be included in the analysis.

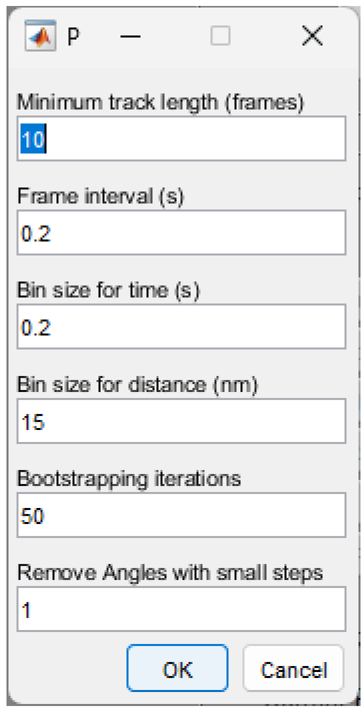

Figure 31: Parameters for angle analysis

After processing the tracks, the Angle analysis window will appear and you will see 3 plots (Figure 32), the left of which is the distribution of angles formed by consecutive positions in all tracked molecules.

The middle, and right-most plots show the temporal and spatial anisotropy, which is defined as the base-two logarithm of the ratio of the number of angles in the forward direction ($-30° \leq \theta \leq 30°$) to the number of angles in the reverse direction ($150° \leq \theta \leq 210°$). To obtain the different timescales, the angles are calculated for integer multiples of a single time-step, i.e. the first value comes from consecutive time-points, the second value comes from every other time-point, etc. The spatial distribution is obtained by calculating the average displacement of the positions at the various timescales, which are then placed in bins of the specified width.

From the GUI, you can copy all of the angles and anisotropy data, copy the individual anisotropy curve data, and save the figure. You can also adjust the displayed plots by adjusting the minimum and maximum Time (or Displacement) and the bin widths. Once finished, click "DONE".

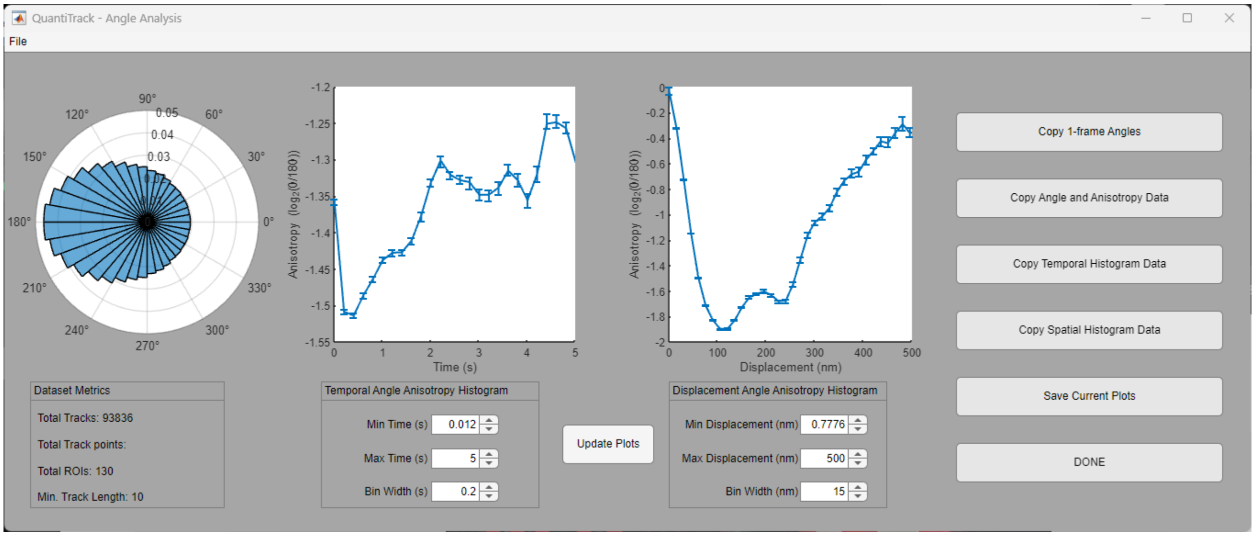

Figure 32: Angle analysis window

Richardson-Lucy (RL) Analysis

Description

The self-part of the jump distance histogram or van Hove correlation (vHc) was calculated as $G_s(r,\tau) = A_s\langle\delta(r_i - |r_{\boldsymbol{i}}(t+\tau) - r_i(t)|\rangle$, where $\boldsymbol{r_i}$ is the position of the ith particle and $A_s = \int d^2r G_s(r,\tau)$ is a normalization constant. As done previously[6,7], we assumed that the vHc can be expanded as a superposition of Gaussian functions $q(r,M) = \left(\frac{1}{\pi M}\right)\exp(-\frac{r^2}{M})$ such that $G_s(r,\tau) = \int P(M,\tau)q(r,M)\,dM$, where M is the mean-squared displacement and $P(M)$ is the probability density of the mean-squared displacements of the ensemble of particles. We then used an iterative scheme developed by Richardson and Lucy and recently adapted to nucleosomes[7] and other transcriptional proteins[6]. From an initial distribution $P^0(M) = \exp(-\frac{M}{M_0})$, the distribution in the $(n+1)^{th}$ iteration, P^{n+1}, can be written as

$$P^{n+1} = P^n \int \frac{G_s(r,\tau)}{G^n{}_s(r,\tau)} q(r,M) d^2\, r$$

Since P^{n+1} is a probability density, we impose the constraint $P^{n+1} > 0$ and that P^{n+1} be normalized.

The resulting distribution of MSDs from tracks obtained from SMT movies can then be classified into different groups. For splitting into bound and diffusing populations, we suggest using a threshold of 0.2 μm^2/s.

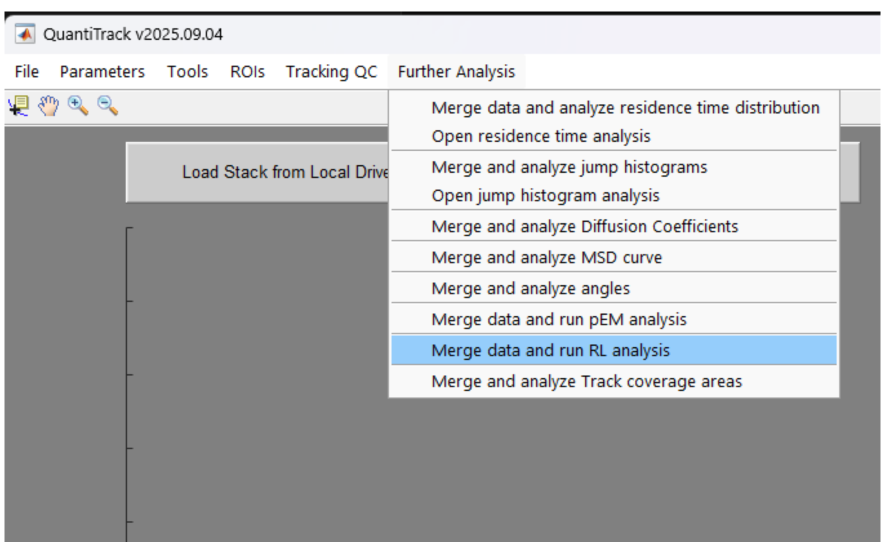

Figure 33: RL analysis location

Procedure

Once you have a trackTable, RLanalysis can be started from the Further Analysis menu, by selecting “Merge data and run RL Analysis (Figure 33). You will be asked to choose the trackTable to be analyzed, followed by some parameters (Figure 34). These parameters are:

2) **Lag time (frames)**: Lag-time (in frame numbers) at which to measure jump distances when computing the vanHove Correlation (vHC) for analysis. We have had good results using a value between 4 and 6 frames.
3) **Number of bins in vHC**: Number of bins to sample the vanHove Correlation for fitting.
4) **Minimum jump distance (um)**: Minimum distance to include in the vanHove Correlation.
5) **Maximum jump distance (um)**: Maximum distance to include in the vanHove Correlation.

6) **Filter out immobile molecules**: Binary flag to tell the analysis whether only diffusing molecules should be analyzed (value = 1) or all data should be included (value = 0).
7) **Maximum D for immobile (um2/s)**: Threshold on the diffusion coefficient to consider a group diffusive. Only necessary if you are filtering out immobile molecules (value = 1 above).
8) **Minimum Track Length**: Removes any tracks from the analysis that contain less than this number of time-points.
9) **Fitting iterations**: Maximum number of iterations to run the Richardson-Lucy algorithm. This should be a large number to ensure convergence.
10) **Min. MSD for plotting**: Minimum MSD value shown in the output plots.
11) **Max. MSD for plotting**: Maximum MSD value shown in the output plots.
12) **Number of MSD bins**: Number of bins to use when plotting the MSD distributions in the output plots.

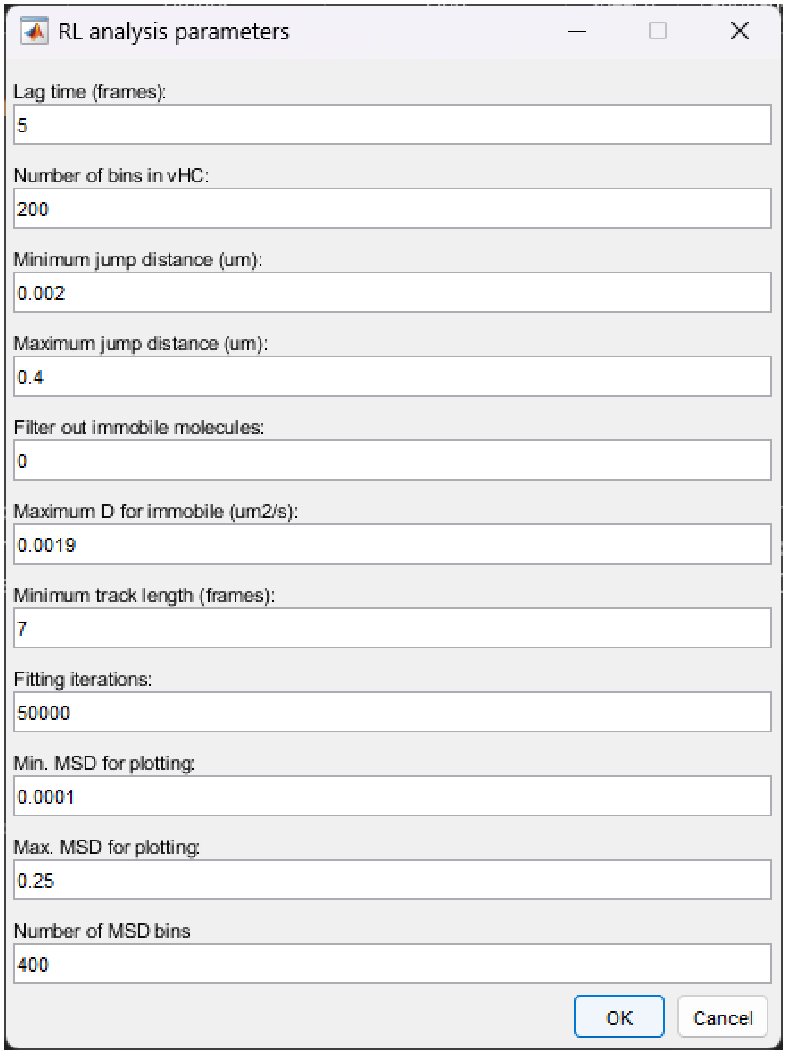

Figure 34: RL analysis parameters

Output

All outputs are automatically saved in the same folder as the 'trackTable' file was located in a folder(s) named 'RL_results_lagtime_LAG_DATESTRING'. Where LAG is the lag-time (in frames) selected in the parameter selection above.

Inside the RL_results folder you will find .mat file that contains the original trackTable and the results of the RL analysis in a table format. There are also several plots that are saved in PNG and EPS format as well as the MATLAB Figure files. These include the final, fitted vanHove Correlation curve, the probability density function for the MSDs used to fit the vHC, and a plot of the classified tracks. In addition, all tracks within each state are compiled into trackTables containing only that state, so that other analyses can be performed on specific states.

Other Tools

Split and Align Colors

For multichannel movies, it may be necessary to perform an alignment based on multi-color beads. QuantiTrack therefore includes a tool to accomplish that task. The tool was developed to deal with TIF-stacks saved from Micro-manager, which contain images in the XYCT format.

Pre-requisites

To perform a channel alignment, we need the following types of images/movies:

1) Multi-channel bead image with 4-8 beads present (a single snap is sufficient, but a z-stack may provide more beads in the field of view).
2) Movies to be aligned

The Fields of view of the above movies should be identical, and the bead image should contain all the channels included in the movie but can contain more channels. With those in hand we can start the Split and Align Color Tool in the Tools menu:

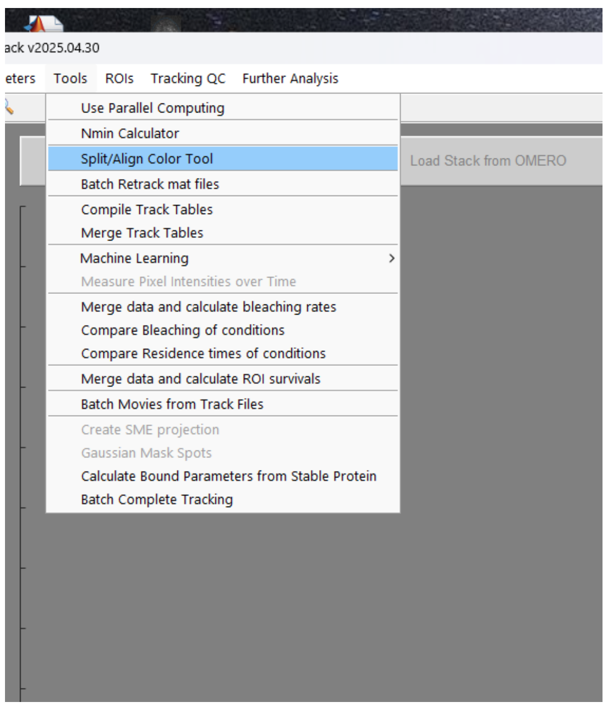

Figure 35: Split/Align Color Tool Location

This will open the Split & Align Channels interface. In this UI you can load in a single image and, optionally, an alignment file. Once a image file is loaded it can be split, and if an alignment file is loaded the channels will be manipulated based on the alignment data using the buttons on the side (Figure 36).

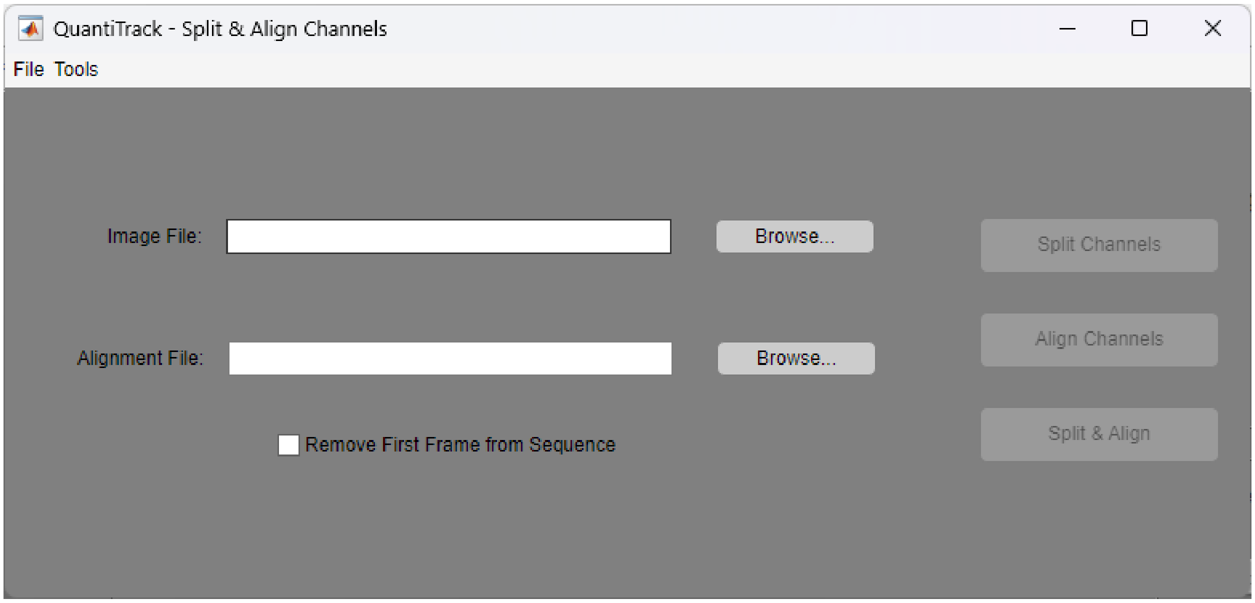

Figure 36: Split & Align Channels tool main interface

Split Channels: a dialog will ask you to specify the channel order in a comma-separated list (NO SPACES), We prefer to use the laser wavelengths to specify channel names. For each specified channel a new tiff stack will be created in the same folder as the selected image file. These files will have the same name as the original with '_CNAME' appended.

Align Channels: applies the transformation indicated in the selected Alignment File (to create this file, see Bead Registration below) to all channels, except for channel 1 which is considered the reference frame. Aligned channels are saved as new tif-stacks with the same name as the original with '_CNAME_aligned' appended.

Split & Align: performs the above 2 steps all at once.

Bead Registration

For channel alignment, we first need to specify the transformation required, which is obtained from the multi-channel bead image. This UI can be found in the Tools menu of the Split & Align Channels window:

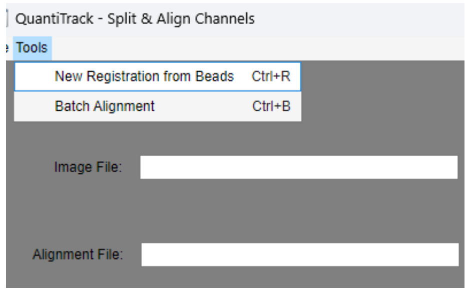

Figure 37:Accessing the Bead Registration tool

Once the Bead Alignment tool is open, the bead image can either be opened from a muti-channel file or individual files containing single channels.

Clicking on any of the Load buttons will open a file-select dialog box where you can choose the tif-stack. When loading the single channel movies, you will be asked what channel name to give this. When loading in a multichannel image, you will be prompted to enter the channel names in a comma-separated format with no spaces. I prefer to name the channels after the laser wavelengths.

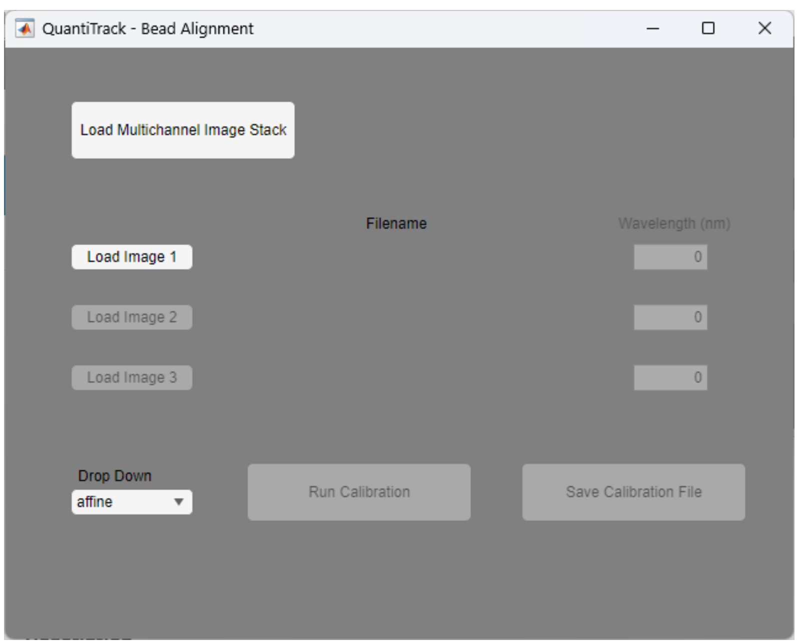

Figure 38: Bead Alignment Window

Once the bead color channels have been loaded and named, we can click on "Run Calibration." This will open MATLAB's built-in tool cpselect, which allows for the selection of control points for image registration. When it first appears, cpselect will show the image specified as Channel 1 on the right (labeled "Fixed Detail") and that specified as Channel 2 on the left (labeled "Moving

Detail”). The upper panel shows a zoomed in region where you can click on the same bead in the two images, and the bottom panel shows the whole image. The rectangle in the bottom panel can be moved to show different areas in the upper panel (Figure 39). After selecting between 4 and 8 beads, the window can be closed with the red X in the upper corner. If a third channel was specified, the cpselect tool will open a second time, with Channel 1 again shown in the right side, and Channel 3 shown on the left. The points selected in the previous step will be plotted. The points on the right do not need to be adjusted, but those on the left likely will need to be adjusted.

After adjusting the points on the left, the window can then be closed. The “Save Calibration File” in the Bead Alignment window (Figure 38) should now be enabled. Click on it and you can choose the location to save the registration data. After saving the Calibration file, you can close the Bead Alignment window.

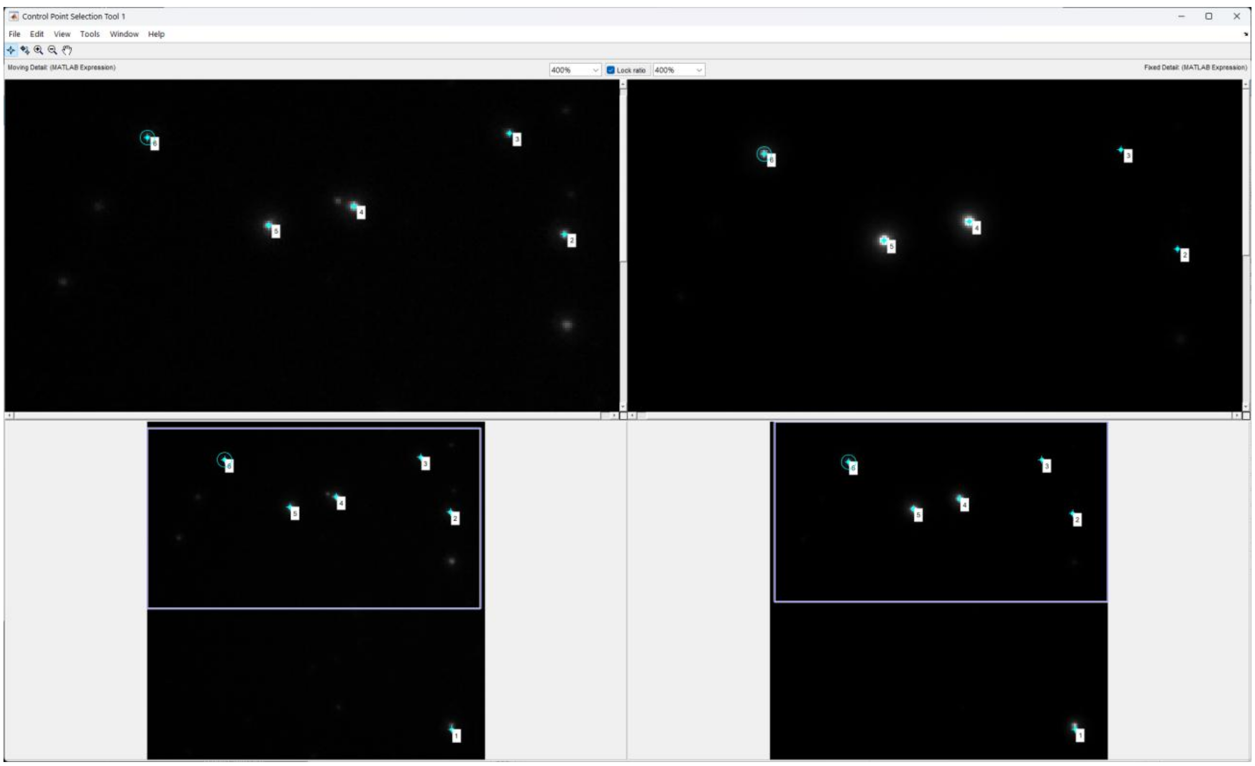

Figure 39:Bead selection

Import ROIs from ImageJ

Regions of Interest (ROIs) created in in ImageJ (https://fiji.sc) can be imported into QuantiTrack, if that is preferred over drawing them within QuantiTrack.

Generate ROIs in ImageJ

With the time-lapse movie open in ImageJ, from the menu bar select Analyze -> Tools -> ROI Manager… (Figure 40).

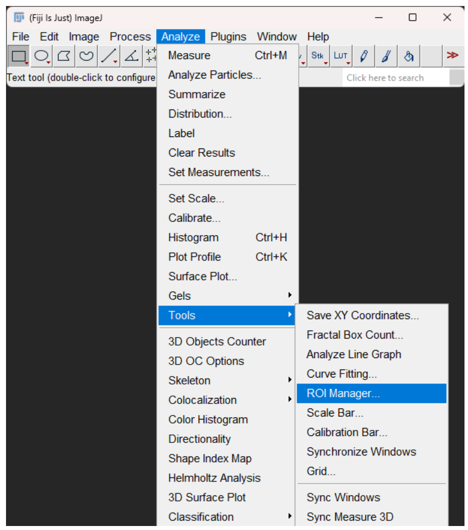

Figure 40: Accessing ImageJ ROI Manager

Create a region with one of the shape tools in ImageJ, and click Add in the ROI Manager(Figure 41)

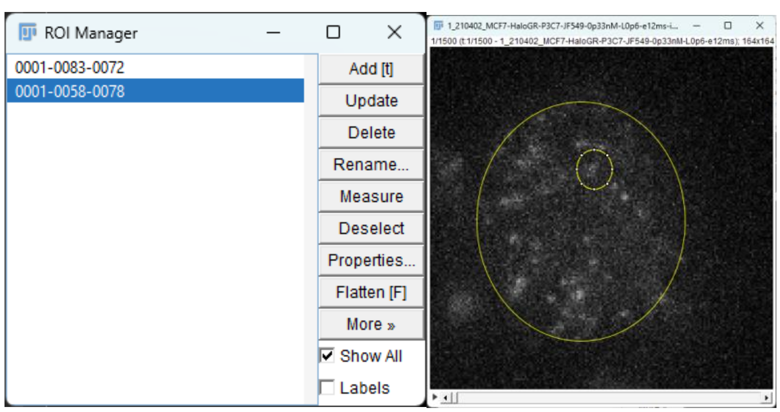

Figure 41: ROI Manager window and Example ROIs created

After drawing all desired ROIs for the movie, make sure all ROIs that you wish to save are highlighted (Shift-, or Control-click to select more than one ROI) and select More >> Save, and choose the save location. This will create a .zip file in the chosen location.

Import ImageJ ROIs into QuantiTrack

After loading in the time-lapse movie into QuantiTrack, From the ROI menu, select 'Import ROIs from ImageJ' (Figure 42). Navigate to the saved location, and Open the ROI.zip file. All ROIs within the file should now be displayed in QuantiTrack, and all should be listed in the ROI list.

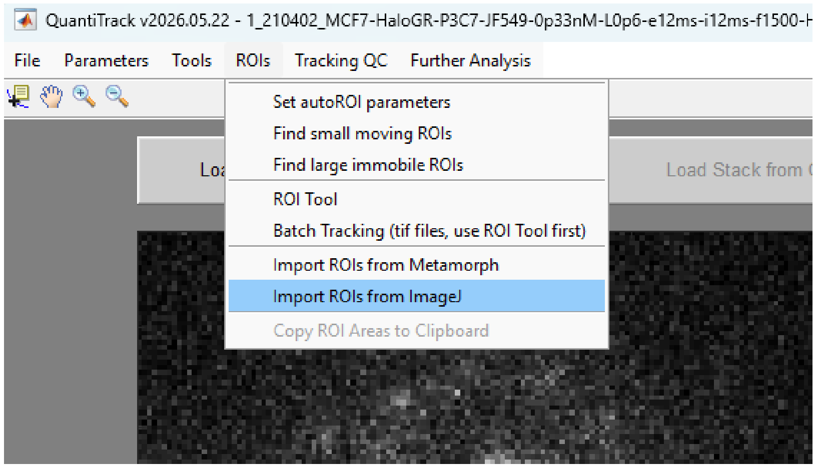

Figure 42: Import ImageJ ROIs

Batch Tracking

When you are confident in the particle identification and tracking parameters, you may want to use these parameters to track a group of movies. QuantiTrack offers a couple of options for performing the tracking protocol on a group of movie files. Because the tracking procedure is centered around regions of interest, the different batch processing options are differentiated by whether the ROIs are manually drawn or automatically found based on intensity thresholding.

Batch Tracking with Manually identified ROIs

Use ROI tool to specify ROIs

We found it cumbersome to load movies into the main QuantiTrack GUI only to specify ROIs, especially if the ROIs were defined by a secondary channel from the main channel used for tracking. We therefore developed a separate panel that is used to draw ROIs, which can be found in the ROIs menu (Figure 43).

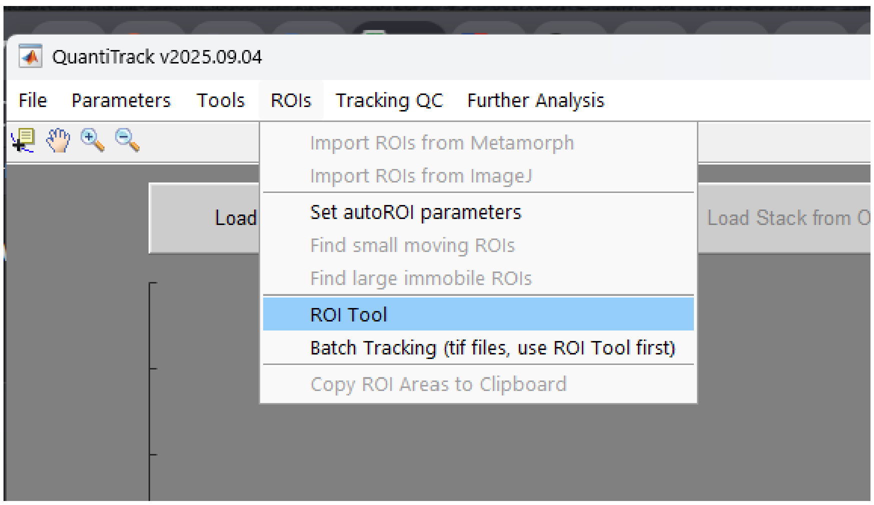

Figure 43: ROI Tool location within ROIs menu

Once opened, the ROI Tool has buttons for loading an Image or movie, drawing ROIs, deleting ROIs, and saving (Figure 44). **Load Image** will open a file browser dialog where you can locate the TIFF file (either single image or a TIFF stack) that you would like to use as a guide for the ROIs. If a stack is specified, you will be asked if you want to display the sum or maximum projection of the full stack. Note, if you choose the full stack, you will need to specify the ROIs in each time point.

Draw ROI allows you to draw a free-hand polygon, where each click will set a vertex, and the ROI is completed when the first point is clicked, or the image is double-clicked. Once the ROI is completed, you will be asked to create a name for the ROI and then be asked if ROIs should be separated into different classifications. The Circle ROI button will generate a circle with the defined diameter centered at the point clicked in the image.

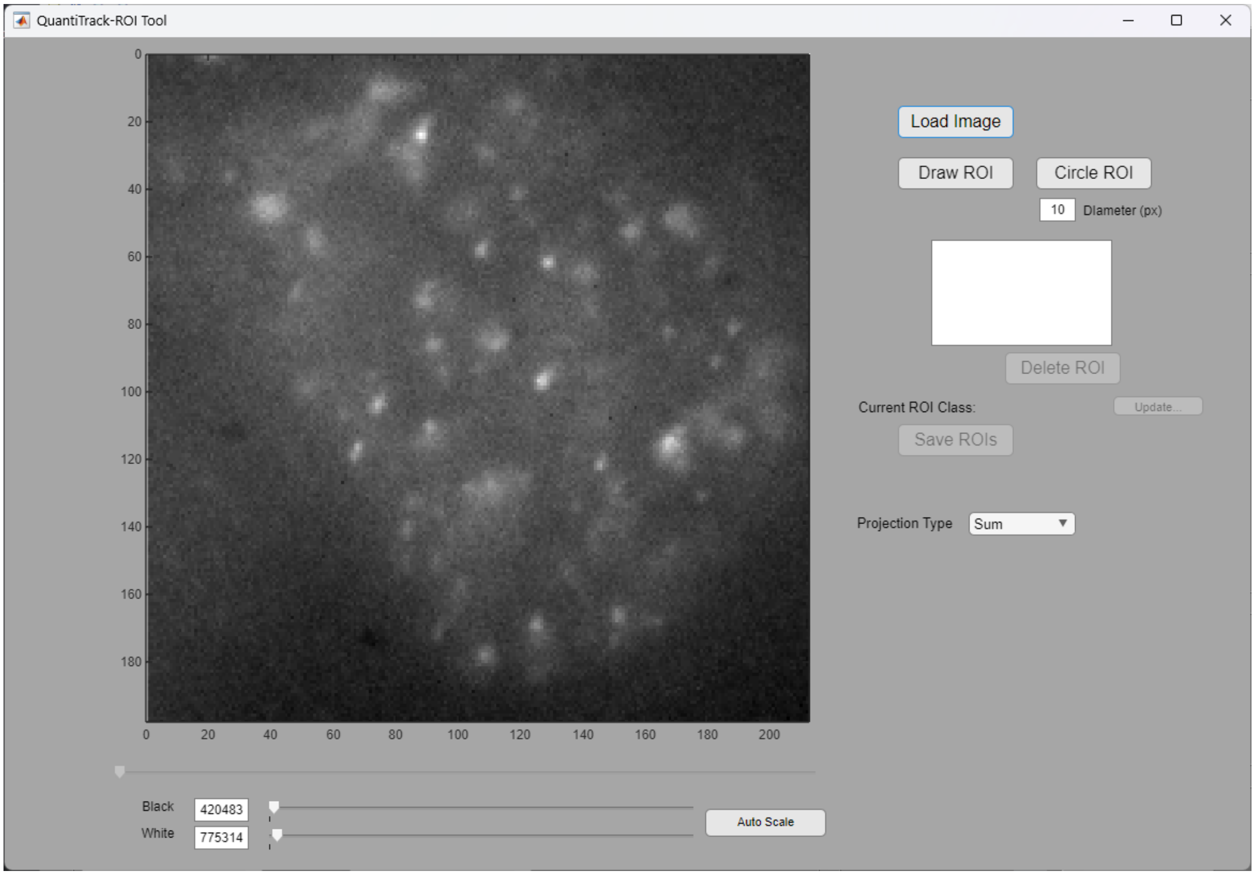

Figure 44: ROI Tool with Sum projected reference image

Once you are happy with your ROIs, clicking on Save ROIs will save a .mat file in the same directory as the image that was loaded, and it will have the same name as the image file, but with the prefix 'mask-'. So, if the image loaded was 'Movie1_488.tif', then the saved ROI file would be called 'mask-Movie1_488.mat'. Note that this may cause issues later if you make ROIs using a secondary reference channel that has a different name from the channel to be tracked.

After clicking the Save ROIs button, you can proceed with loading the next movie, and repeat the process until you have ROIs for all of the movies that you want to track.

Rename ROI files, if necessary

If the ROIs were specified from a secondary reference channel, the names will need to be changed to the name of the movie that is to be tracked.

For example, you have a 2-color, time-lapse movie that was split and aligned (see Split and Align Colors), resulting in the movie that will be tracked: ‘Movie1_647.tif’; and a reference movie: ‘Movie1_488.tif’, which was used to create the ROIs, resulting in the creation of the file ‘mask-Movie1_488.mat’. In order for the batch processing to work, you will need to change the name to ‘mask-Movie1_647.mat’.

Run Batch Processing

Now you can run the Batch Processing from the ROIs menu (Figure 45). When you click on the Batch Tracking, a dialog will pop up asking to specify all of the parameters. These are the same parameters specified in the main QuantiTrack GUI during processing. You will then be asked to select the parent folder that contains all of the movies to be processed (and their corresponding ROI files). The program will search through all subfolders of the selected directory for ROI files contained within.

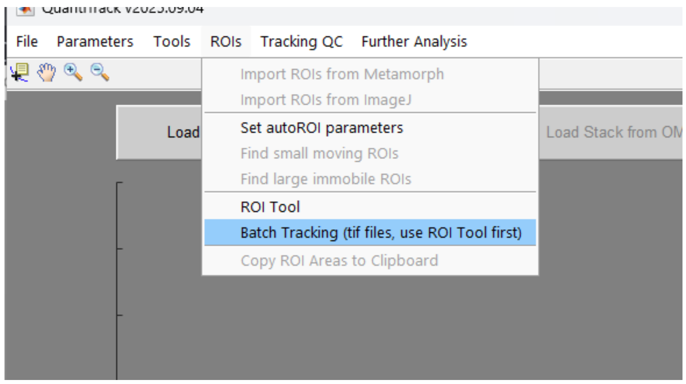

Figure 45: Batch Processing with manually defined ROIs

You will then be asked to specify the location in the Track Database where the tracked files will be saved. Then you can choose the type of filtering, and the tracking algorithm, and batch processing will begin.

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